

Lectures

on

“Philosophical Propaedeutics”

from the Academic Year 1860–61

translated from the Danish by H. Halvorson

R. Nielsen

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Translator's Note on Terminology

Nielsen lectured in 1860–61 in a philosophical Danish that had been formed on German models, and much of his vocabulary is a set of Danish calques on Kantian and Hegelian terms. A few of these words have modern English cognates that mean something quite different, and a reader who takes them at face value will be misled. The renderings below are used consistently throughout; a dagger footnote marks the first occurrence of each in the text.

***den sunde Forstand* — “sound understanding”**

Literally “the healthy understanding”; the standing Danish equivalent of German *der gesunde Menschenverstand*, and close to English “common sense.” But *Forstand* is *Verstand*, the understanding as opposed to reason, and Nielsen’s whole argument turns on its being a *faculty* that can be shown to contradict itself — not a stock of shared opinion. He almost always sets it in quotation marks, and those are preserved. The fuller form *den sunde Menneskeforstand* is given as “sound human understanding.”

***Anskuelse* — “intuition”**

Anschauung: the immediate presentation of an object to sense, in Kant’s technical sense. Never “opinion” or “view,” which is what the cognate suggests in modern Danish.

***Forestilling* — “representation”**

Vorstellung: the mental image or presentation, standing between sensation and concept. Nielsen’s argument at pp. 10–11 depends on the contrast between what can be *thought* and what can be *represented* — a round square can be thought, he says, but not represented.

***sandselig, Sandsning, Sandseindtryk* — “sensuous, sensation, sense-impression”**

sinnlich: pertaining to the senses. “Sensuous” here carries none of the modern connotation of “sensual.”

***Skarpsindighed / Spidsfindighed* – “acumen” / “over-subtlety”**

A matched pair in Danish (*skarp* = sharp, *spids* = pointed) that English cannot match morphologically. The first is the virtue of exact analysis; the second is its vice — subtlety that dazzles by distinctions only half true. Distinguishing them is the business of Part I.A, and Nielsen insists the same utterance may be either.

***Menen / Viden* – “opining” / “knowing”**

The Platonic *δόξα/ἐπιστήμη* distinction. Nielsen puns on it at p. 22 (“the question is not of opining but of knowing — still, what is your opinion?”).

***Ophæve, Ophævelse* – “sublate, sublation”**

aufheben: at once to cancel, to preserve and to raise up. Where the context is plainly non-technical the ordinary “cancel” or “annul” is used instead.

***Tvetydig / Tvesidig* – “ambiguous” / “two-sided”**

Another pair the English breaks: literally “two-meaning” and “two-sided.” Nielsen’s definition of dialectic at p. 15 is that it is the science which discovers the *two-sided* in the *two-meaning*.

***Grund / Følge* – “ground” / “consequent”**

Grund/Folge, the logical pair, not cause and effect.

***dialektisere* – “dialecticize”**

Nielsen’s own verb, and deliberately ungainly in Danish too: to derive one thing from another by dialectic. It is kept rather than paraphrased, since the dialogue of pp. 24–28 turns on the question of what can and cannot be so derived.

***Grundsætning / Sætning* – “fundamental principle” / “proposition”**

Sætning does duty for both “sentence,” “proposition” and “theorem”; where Nielsen means one of the laws of thought, “principle” is used.

diversi respectus

Left in Latin, as Nielsen leaves it: the scholastic maxim *diversi respectus tollunt omnem contradictionem* — different respects remove every contradiction. It is the hinge of the first dialogue, and translating it would blunt the joke that B. has never troubled to study what a *respectus* is.

Nielsen cites three of his own earlier books by short title, without further identification: *Propæd.* or *Propædeut.* is *Philosophisk Propædeutik i Grundtræk* (Copenhagen 1857); *Philos. og Mathem.* is *Philosophie og Mathematik*; *Mathem. og Dialekt.* is *Mathematik og*

Dialektik. These are left in Danish, as he gives them. All page references in such citations are to the Danish originals.

Nielsen's own footnotes are marked, as in the print, *) and **); translator's notes are marked with a dagger and open with the words *Translator's note*.

Preface

Should the present work attract any attention, the interest taken in it will doubtless be of two kinds. With the question of the possible worth of its contents, most readers, keeping to the letter of the title, will surely join a second and more practical question: *on what principles, and to what extent, was the philosophical propaedeutic here delivered to the students?*

As regards this latter question, it will not be enough to remark merely that the motives, leading ideas and whole design of the present exposition correspond to the oral lectures, but that in matters of detail a good deal has been altered, something added, something else omitted, and so forth; the reader will think himself entitled to a clearer account.

It is therefore expressly noted that, apart from a few less considerable editorial changes, the whole of the first part, pp. 1–30, is a straightforward reproduction of the lectures as delivered. The same holds of the introductory piece of the second part, *General Remarks*, pp. 30–39; of Philosophy's Relation to Mathematics, pp. 39–63; of Philosophy's Relation to Chemistry, pp. 81–100 and pp. 135–177; and of Philosophy's Relation to Physiology, pp. 177–180, pp. 199–288, pp. 386–437, pp. 475–482.

In the presentation of the remaining portions, by contrast, considerable changes have been made. The section on the psychophysical investigations was not touched upon at all in the oral lectures; sharper and further-reaching analyses have been introduced here and there; and it goes without saying that by far the greater part of the contents of the notes was added in the course of the revision.

Since the effort to give the treatment of the physiological problems in particular that fullness which investigations of so intricate a nature can hardly do without brought with it a considerable expansion of the second part, the third part — Philosophy and Academic Study — has been confined to a bare statement of its moments, and is appended as a separate part only in order to exhibit the course of the whole.

11 June 1862.

R. Nielsen.

I.

The Concept of Philosophy.

A.

The Doctrine of Contradiction.

What is philosophy? “A science full of contradictions!” However much this definition may seem, viewed from one side, designed to pass sentence on philosophy altogether, it is nonetheless well founded, both in the history of the science and in the nature of the case. For any other science such an admission would be annihilating; for philosophy it furnishes an essential starting point, a dependable fulcrum, a firm and unshakable foundation. That philosophy must contain contradictions is a matter of course, since it is precisely the science which everywhere discovers and exhibits contradictions — in existence itself no less than in sound understanding’s[†] grasp of existence (Propæd. pp. 40–44)[†] — and it is just in this respect that philosophy is to be understood as a *general doctrine of contradiction*.

As a doctrine of contradiction, however, philosophy must not merely discover and exhibit contradictions; it must also test them, for there are differences among contradictions. Some are superficial, unnecessary, easily avoided; others are insistent and not to be waved aside. Some creep in through thoughtlessness, others are manufactured by over-subtlety, others again brought to light by acumen.[†] Some contradictions play about

[†] *Translator’s note.* *Den sunde Forstand*: the Danish equivalent of German *der gesunde Menschenverstand*, and close to English “common sense.” But *Forstand* is *Verstand*, the understanding as against reason, and Nielsen’s argument turns on its being a faculty that can be shown to contradict itself. He sets it in quotation marks almost everywhere, and those are kept. See the Translator’s Note on Terminology.

[†] *Translator’s note.* Nielsen’s short titles for his own earlier books, cited throughout without further identification: *Propæd.* (also *Propædeut.*) is his *Philosophisk Propædeutik i Grundtræk* (Copenhagen 1857); *Philos. og Mathem.* is *Philosophie og Mathematik*; *Mathem. og Dialekt.* is *Mathematik og Dialektik*. Page numbers refer to the Danish originals.

[†] *Translator’s note.* *Spidsfindighed* and *Skarpsindighed* form a matched pair in Danish (*spids* = pointed, *skarp* = sharp) that English cannot match. The first names the vice of subtlety that dazzles by half-true

as in a game of thought; others mark essential knots, difficult problems not easily untied.

The following examples may serve to show how necessary it is, where contradictions are concerned, to distinguish the superficial from the essential, the acute from the over-subtle.

The Liar (ψευδόμενος). This contradiction is put as follows: when someone says that he is lying, is he lying in that very statement, or is he telling the truth? The answer demanded is either yes or no. "Yes, he is lying!" — And he says so himself, calls the thing by its right name, says what is true, and hence is not lying. "No, he is not lying, he is speaking the truth!" — Then one must believe what he says; but what he says is precisely that he is lying; and if it is true that he is lying, then he is not speaking the truth.

That there is a contradiction here, sound understanding can readily grasp; but what sort of contradiction is it, and where does the fault lie?

Equal Magnitudes. That a magnitude is equal to itself, and that two, three .. magnitudes are equal to one another — in this sound understanding sees no contradiction. But if it were asserted that two magnitudes cannot possibly be equal to one and the same third, and hence equal to each other, because that would be a contradiction, sound understanding would feel itself affronted. Is there then really any contradiction whatever in the proposition just cited? To see what is actually being spoken of here, one must expressly note that the talk is not of things that have magnitude but of magnitudes. If the talk were not of magnitudes themselves but of things that have magnitude — an apple, say, a triangle, an hour, and so on — there would certainly not be a trace of contradiction. That two triangles, each of which is equal to one and the same third, are equal to one another, goes without saying. So long as we have to do with concrete magnitudes, difference and plurality can coexist perfectly well with equality; but when it is pure magnitudes that are in question, the matter stands otherwise. A plurality of magnitudes necessarily presupposes that the one can be distinguished from the other: without mutual difference, no plurality. But by what should pure magnitudes be distinguished from one another if not by size? If the magnitudes are equal to one another, then every glimmer of difference has vanished, and with it the plurality: to say that several magnitudes are equal to one another is therefore to say, in other words, that they are not several but one and the same magnitude, which, however many times it be repeated, is not thereby multiplied but remains the very same.

Should "sound understanding" wish to reject this consequence because there is something offensive about it, that can hardly be done out of hand by a mere dictum of

distinctions; the second the virtue of exact analysis. Distinguishing them is the business of this section, and Nielsen insists that one and the same utterance may be either.

authority; a sensible rejection must set about showing wherein the difficulty consists and how it is to be removed. If one and the same magnitude can, by being named twenty times, become twenty magnitudes equal to one another, then one and the same species must likewise be able, by being named twenty times, to become twenty species; but if it is not enough to repeat the same magnitude — if in repeating it a difference must also be thought — then where is that difference to come from?

The Instant.[†] If the old question — when did Socrates die? — is answered by pointing to the year of his death, sound understanding will hardly suspect any contradiction in that. Unless indeed it be that Socrates has no year of death at all, inasmuch as he did not take a year over dying: he was hale and healthy through the whole of that last year, until he drank from the cup. But if, in accordance with a well-known and warranted usage, one means by his year of death the year in which the day of his death fell, then there is no contradiction here. If it be objected that he did not take a whole day over dying either — well then, let one think of the hour of his death; and if one will go to the very limit, of the instant of his death. So then: when did Socrates die? “At the instant he ceased to live!” But this answer, which tells us only that he died when he died, is no answer at all. The question concerns the point of time in Socrates’ existence at which his death occurred. To specify this point of time, four cases may be conceived: either he died α) while he was still living, or β) when he was already dead, or γ) when he was partly living and partly dead, or δ) when he was neither living nor dead. But α) to die while one is still living, to be a living dead man, is surely a contradiction. And to die when β) one is already dead is surely a contradiction too. The instant at which Socrates died must therefore have been the instant when, lying on the border between life and death, he was in such a condition that one part of him — his feet, say — was already dead while another part was still alive. By this explanation, however, the question is not answered but only pushed further back; for the question then is: when did the already dead part die? If the answer is that the already dead part died when it was still partly alive and partly dead, then the contradiction concealed in case γ) becomes obvious. And to conceive Socrates in a condition in which he was neither living nor dead is, finally, likewise to conceive a contradiction (δ); for how long does a living thing live? Until it dies!

“A philosophy that stands in open contradiction with ‘sound understanding’ is false; a philosophy that conforms itself immediately to sound understanding is superficial. Philosophy does not wish arbitrarily to contradict the natural understanding, but only to show

[†] Translator’s note. *Øieblikket*: literally “the eye-blink,” the smallest unit of time. “Instant” is used rather than “moment” throughout, to keep the word clear of the Hegelian *Moment* that Nielsen also employs.

that so-called sound sense, without having any inkling of it, incessantly contradicts itself.” (Propæd. p. 41).

In so far as contradictions of these various kinds develop in the course of thinking, and their origin is viewed from the subjective side, the psychological share that reflection has in the contradictions offensive to “sound sense” must be determined more closely. The aberrations belonging here may be briefly characterized as the *eristic*, the *over-subtle* and the *sophistical*.

The striking mark of eristic is a one-sided bent for quarrel and strife, a delight in shining everywhere by objections, in raising difficulties, in tearing down what is generally accepted and stubbornly defending one’s own divergent claims. As for ancient philosophy, it is above all the Megarian school, akin to the Eleatic, that has earned the name of eristic; but neither can it be denied that every distinctively developed form of philosophy has its polemical element, and that eristic, for all its dark sides, is to that extent of unmistakable importance in the development of the science. For insight and knowledge it is not enough that one form what is commonly called a reasonable opinion about things, and, by combining sense-impressions with natural judgment, be led to distinguish correct from incorrect, acceptable from objectionable opinions; by means of opinions one attains only the probable, not the true. Whoever would rise from mere opining to knowing[†] must arm himself against the deceptions of sense, let thought test what has been accepted, and acknowledge as true only what admits of being thought — declaring untrue, without regard to the multitude’s otherwise unimpeachable opinions, everything that does not admit of being thought. That philosophy, on these terms, must come into manifold conflict with sound human understanding is evident. (Cf. Philos. og Mathem. p. 12, note; Mathem. og Dialektik pp. 34–35, pp. 116–117, pp. 131–32, etc.) Rigorous analysis of concepts demands acumen; but acumen that is one-sided and unfruitful, dazzling by distinctions which are subtle yet only half true or even untrue, yields over-subtlety. Easy as it is to state the difference between acumen and over-subtlety so long as one keeps to generalities, it is just as impossible to lay down a rule valid for every particular case; for one and the same utterance may, seen from one side, look over-subtle, and yet, seen from another, bear the stamp of sound and genuine acumen. The contradiction set up in “the Liar” may serve as an example. If the task is to drive home the truth that one should not undertake off-hand to answer every question, since there are questions whose apparently clear and easily grasped sense proves on closer inspection to be senseless, then “the Liar” may stand as a model of

[†] Translator’s note. Menen and Viden, the Platonic pair δόξα/ἐπιστήμη. Nielsen returns to the pun at p. 22.

genuine acumen. If, on the other hand, one guilelessly takes up the answer demanded — and does so in such a way that, watching with astonishment as the affirmation turns into a denial and the denial back into an affirmation, one ends by imagining oneself before some deep riddle, some exceedingly intricate knot whose solution may perhaps shed a hitherto unknown light upon “the eternal truths” — then over-subtlety has outwitted sound understanding and caught it in its web.

In secret league with over-subtlety and hair-splitting we find sophistry. The sophist is not content to exhibit contradictions; he manufactures contradictions, plays with contradictions, draws inferences, and dazzles by false inferences. For it is one thing to bring contradictions into relief, another to misuse them in the service of sophistic; how great the difference is may be seen, for instance, from what was said above about “the Instant.” No sensible person can regard it as sophistry to call attention to the contradictions upon which thought must necessarily strike when it tries to give an account of the meaning of the instant; only when it is a question of the use made of the contradictions does sophistry become recognizable, by what is false, untenable and dazzling in the inferences. Thus if anyone were to infer, from the contradictions contained in the thought of the instant of death, the unthinkability of death, and from the unthinkability of death its impossibility, and were then to build upon that a new proof of the immortality of the soul, fitted out with a peculiar acumen, we should have to consider him a sophist.

B.

The Doctrine of Possibility.

If philosophy as a doctrine of contradiction is not to dissolve into over-subtlety and sophistry, it must always find itself able, from one side or another, to master the contradictions. It must then be able to determine their relation both to things and to our reflections upon things, both to objective being and to subjective thinking; in seeing through the essence of contradiction it must discover the marks of what is in itself thinkable and what is in itself unthinkable, and thereby, so to speak, wring from contradiction a dependable measure of the acceptable and the objectionable, the reasonable and the absurd, the possible and the impossible: a thoroughgoing doctrine of contradiction is inseparable from a *general doctrine of possibility*.

What, then, is philosophy? “A doctrine of the possible.” But since philosophy cannot possibly be a doctrine of everything possible, this Wolffian definition is satisfactory

only on condition that the concept of possibility — its extension, its content, and its relation to contradiction — be exactly determined. The principle that everything is possible which does not contain a contradiction (*possibile est, quod non implicat contradictionem*) — which in turn presupposes that everything containing a contradiction is impossible — will, on the strength of what has gone before, hardly furnish a dependable guiding line, whether dogmatic or critical.

In the first place it is self-evident that the principle as stated specifies possibility only formally, and is therefore unfit to orient us with respect to the real possibilities. Even if for the time being we declare everything that contains a contradiction impossible, and everything that does not contain a contradiction possible, the concept of possibility so given will still be so negative, so empty and unsettled, that it cannot be brought into any positive connection whatever with the actual. To determine the possibilities that hold in actuality, it is necessary to know actual things together with their conditions and laws; and knowledge of actual things is reached only by the road of experience. Without the correctives of experience, dogmatizing thought with its empty concepts will be as readily tempted to set the real possibilities too narrow bounds as to allow them too wide a play. Or who, back in the days when Wolff was fixing the bounds and defining the possibilities, would have seen the possibility that ships should be able to go both against wind and against current; that carriages without horses should be able to run — not like horses, but swift as flying birds; or the possibility that a man in London should be able to correspond with a man in Calcutta as quickly and easily as if they were talking together? So long as philosophy, with its formal concepts and definitions, attempts to exhaust existence and to set itself up as judge over what is possible and impossible in actuality, so long will sound human understanding have manifold occasion for the triumphant application of Hamlet's words: *There are more things in heaven and earth, Horatio, than are dreamt of in your philosophy.*

But even if philosophy gives up trying to fathom the positive, empirical content of the real possibilities, it may still contribute essentially to purifying and heightening our knowledge of the possible, provided only that it can secure for us the negative side of possibility by determining the impossible. Now if the impossible can be known only as what is in itself unthinkable, and if the unthinkable must always be recognizable by contradiction, then the fundamental question presses itself upon us of its own accord: *how, then, does the contradictory stand to the thinkable?* Here we are at a decisive turning point.

In presupposing without more ado the unity of thinking and being, dogmatic phi-

losophy proceeds at the same stroke from the assumption that the contradictory is unconditionally an unthinkable, and the unthinkable an impossible. But is the contradictory really unthinkable in every respect?

“What does contradiction mean? An existing nothing is a contradiction; a round square is a contradiction; a flying horse is a contradiction too. And yet these contradictions are of different kinds.” (Propæd. p. 43).

If what is contradictory in itself is straightway unthinkable, how then does thought come to apprehend this unthinkable? How does it happen that we not only understand what is meant by the word “contradiction,” but also see clearly what lies in the essence of contradiction? If we set, say, “something” against “nothing,” “that which is” against “that which is not,” thought finds it just as easy to grasp the negative as to grasp the positive; and if we set the proposition “that which is, is” against the proposition “that which is, is not,” one notices at once that a conflict between subject and predicate is as evident to thought as an agreement. But if thought apprehends the conflict as readily as the agreement, sees through an existing nothing as clearly as an existing something, and so far finds the former as thinkable as the latter — how then can what is in itself conflicting and contradictory show itself to thought as an unthinkable, and this unthinkable coincide with the impossible?

To extricate oneself from the difficulty, let one try letting representation come to the support of thinking.[†] Pure, abstract thinking, detached from all intuition and representation, does indeed grasp the negative as readily as the positive, since the negative is precisely abstraction’s own abstract product; but when the light of intuition dawns, when representation joins in, the opposition between the positive and the negative, between something and nothing, becomes conspicuous. For something admits of being intuited, whereas nothing, pure nothing, does not; to the positive there answers a representation, but nothing — the negative — repels every representation, and is for that reason in turn repelled and rejected by representation.

How thinking, by being combined with representation, comes to apprehend the contradictory as the unthinkable, and the unthinkable as the impossible, may be seen from the example of the round square. The proposition “the square is round” can indeed be thought, in so far as the predicate’s conflict with the subject can be thought; but the round square cannot be represented, since the representation of the round displaces

[†] *Translator’s note.* *Foretelling* is *Vorstellung*, the mental presentation or image standing between sensation and concept; *Anskuelse*, just below, is *Anschauung*, intuition in Kant’s technical sense — never “opinion” or “view.” The whole argument of this page turns on the contrast between what can be *thought* and what can be *represented*.

and effaces the representation of the square. Thought, seeing that the round square makes a demand upon representation which representation must reject, now brands the contradiction as something objectionable, declares the round square an unthinkable, and this unthinkable an impossible. That it is precisely intuition and representation which in such cases must help thought to discover the unthinkable is seen not only from propositions in whose components representation manifestly rules — "black is not white," "thick is not thin" — but also from examples that do not directly display the contradiction in their words and manner of expression, as when one asks: is a regular hexahedron possible?[†] is it possible for a line parallel to an ellipse to be an ellipse? and so on.

If only we could rely on intuition and representation to furnish, always and in every case, infallible marks of the thinkable and the unthinkable, we should at least have something firm to hold to here; but now look at the example of the flying horse! Natural sense surely feels that under the name "flying horse" a contradiction lies hidden. But to whom is sound human understanding now to turn to have the contradiction cleared up? To thinking! That is no help: pure thinking thinks everything, the unthinkable included. To intuition! No help either: intuition lives with the poet, and the poet rides Pegasus, and Pegasus reconciles intuition with every winged horse. To a union of thinking and intuition! But how can a union of two unreliaables yield anything reliable? To sense-impressions! Sense-impressions can indeed, in union with thinking, teach us something about the possible, in so far as they teach us something about the actual, of course; but what do they teach us about the merely possible, and what could they possibly teach us about what is in itself contradictory and impossible?

To determine the relation of possibility to contradiction, it is above all necessary to grasp *the principle of contradiction* — its form, its validity, its application.

In formal logic we find the propositions of identity and of contradiction set out so separately that we are thereby supplied not merely with two fundamental principles but, into the bargain, with two independent principles. If there is really anything in this duality of principles, we should expect some account of the difference of principle between them. But wherein is the difference supposed to consist? To explain the proposition of identity — *A is what A is* — one slips in underneath it the proposition of contradiction: that *A is what A is* precisely because *A is not other*, and cannot be other, than what *A is*. Now if the principle of contradiction has had to do service in setting

[†] *Translator's note.* The example is puzzling as it stands, since the regular hexahedron is simply the cube. Nielsen's point is presumably that these are questions whose answer cannot be read off the words — unlike "black is not white" — so that intuition must be consulted before one can say whether a contradiction is present at all.

up pure identity — and that, in the logical sense, before it was even born — then it is no more than fair that the principle of identity, once born and the logical birth-pangs happily over, should in its turn help the principle of contradiction to due recognition. Why, then, is no thing what it is not? Answer: because every thing is what it is. And why is every thing what it is? Answer: because no thing is what it is not. In other words: when identity is asked after, one answers with contradiction; and when contradiction is asked after, one answers with identity. On closer inspection it must undeniably strike sound human understanding that there are here not really two fundamental principles but only one.

Yet suppose that, in order so far as possible to avoid conflict with traditional forms, we let formal logic keep its two fundamental principles undiminished — so that, for the sake of “certainty” and “truth” and “determination,” “statuation,” “whatness,” “decision,” “unanimity” and “concordance,”[†] — we held identity literally apart for itself and contradiction apart for itself: would such a holding-apart then perhaps help sound understanding to a clear and distinct insight into the relation between the contradictory and the impossible? At first glance it seems that some difficulty still remains. For if every *A*, provided only that it is what it is, has by the principle of identity a claim to “a valid is,” then it looks almost as though the senseless and the thoughtless could likewise make a claim — a valid claim — to “a valid is”; if every thing, provided only that it is not what it is not, can satisfy the principle of unanimity and concordance, why should the conflicting, the absurd, the self-contradictory and the foolish not be able to satisfy the principle of unanimity and concordance as well?

But to overlook the principle of contradiction, still less to reject it, will not do at all; on the contrary, when thought attempts to deny the validity of the principle of contradiction, it is led by the very denial to confirm it. To posit “black is white” instead of “black is not white” is not only possible but reasonable and necessary if the corresponding contradiction is to be made properly plain; yet in being branded in this way as a contradiction, the contradiction vindicates the principle. The unthinkable, in this sense, is not a something that thought simply cannot think; the unthinkable is the thought which thinking itself condemns, in view of a purpose given within a given context — a thought that comes into being only to vanish, a thought in whose sublation[†]

[†] *Translator’s note.* The list is ironic: Nielsen is parading the scholastic-sounding coinages under which formal logic defends its two principles. *Hvadhed* is a Danish calque of *quidditas*, “whatness”; *Statuering* is the laying down of something as established; *Eenstemmighed* and *Samstemmighed* are “unanimity” and “concordance.”

[†] *Translator’s note.* *Ophævelse*, Hegel’s *Aufhebung*: at once cancelling, preserving and raising up. Where the context is plainly non-technical, the ordinary “cancel” is used instead.

thinking recognizes its own law and satisfies itself. Only when the unthinkable is thus determinately set against the thinkable can the latter be called the logical mark of the possible and the former that of the impossible. Contradiction, then, logically regarded, cannot be abolished outright. Could the principle of contradiction abolish contradiction once and for all, it would at the same stroke render itself superfluous. But since thought cannot stir without contradiction — its sting and its spur — stirring too, the principle stands in force because the cooperation of contradiction stands in force; in order to determine the thinkable, sift the possible and dismiss the impossible, the principle of contradiction must incessantly contradict contradiction, must in resolving contradiction uphold contradiction.

In the application of the principle of contradiction the difference between critical and dogmatic philosophy comes into view. Agreeing with dogmatism in setting contradiction outwardly hostile against thought and the movement of thought, critical philosophy then diverges from the dogmatic in this: that in the unthinkable it sees only the subjective side of the impossible, and not its objective side as well. When Kant attempts to think the origin of the world, he strikes, quite rightly, upon contradiction; the same recurs when he tries to think the essence of substantiality, of necessity and of freedom objectively. The acute thinker, consistent from his own point of view, is here so far from being rid of the contradictory that on the contrary, seeing contradiction tie its knots and settle into antinomies that cannot be waved aside (Propæd. pp. 29–30; Mathem. og Dialekt. pp. 125–126), he declares it insuperable. The objective essence of the world belongs therefore, according to Kant, to what is for us unthinkable; but what does Kant infer from this? By no means that the reality of an objective world should on that ground be held impossible — that is an inference which first arises with Fichte (Propæd. p. 33) — no, the author of "Kritik der reinen Vernunft" warns precisely against precipitate inferences from the subjectively unthinkable to the objectively impossible. The separation of thinking and being, the original characteristic mark of critical philosophy, rests precisely on the presupposition that for the human understanding there are essential, necessary, insoluble contradictions.

But are the Kantian antinomies really insoluble? That depends on whether one chooses to grant contradiction a merely subjective significance, or a significance at once subjective and objective. Such a choice cannot of course be arbitrary; it must follow from the nature of the case. If, then, it turns out that by denying the objective validity of contradiction, holding fast to the critical separation of thinking and being, and so rejecting every attempt to know "*the thing in itself*," one by no means escapes

contradiction but rather entangles oneself in new and greater contradictions — what then?

C.

Dialectic.

What, then, is philosophy? “A doctrine of ambiguity, a science of the ambiguous.” The correctness of this definition seems at least to find ample confirmation in what has gone before. For although we did distinguish between acumen and over-subtlety, we had to admit that one and the same thing could be taken both as over-subtle and as not over-subtle; when it was a matter of separating the thinkable from the unthinkable, what came out was that the unthinkable was something that both could and yet could not be thought; and when it finally came to a resolution of contradiction, the upshot was that contradiction both could and could not be resolved. Should sound understanding find such a predilection for the ambiguous suspicious, let it be noted that the expression “doctrine of ambiguity” is itself ambiguous. While higher instruction in making the clear obscure, the distinct veiled and the intelligible unintelligible is always to be sought among the sophists, philosophy must contrive to set the ambiguous in another light — a light that makes the ambiguity vanish. As the science which everywhere discovers the two-sided in the ambiguous,[†] and by means of contradiction clarifies the doubling, philosophy is *dialectic*^{*)}.

The truth that philosophy by its higher nature is and must be dialectic was categorically stated by Plato. (Cf. *Propædeut.* pp. 124–129; p. 150 and elsewhere.) In Plato the presentation of the dialectical (διαλεκτική from διαλέγεσθαι) has, as is well known, taken the form of conversation; and connected with this is the fact that the dialectician is characterized precisely as the one who in conversation knows how to ask and to answer

[†]Translator's note. *Tvetydig* and *tvesidig*: literally “two-meaninged” and “two-sided.” The definition depends on the Danish pair, which English cannot reproduce.

^{*)}That the form of philosophy is in its essence dialectical must not be misunderstood as though the development of concepts had everywhere to display immediately the formal antinomies by which logical transitions are motivated; on the contrary! Philosophical exposition can and should also take up reporting, descriptive and critical components. The only decisive point is that the elements communicated in the form of report, description, critical discussion and the like should everywhere show themselves, by the design of the whole and the prevailing current of thought, to serve the dialectical. There is in the Hegelian method a formal overstrain, a logical over-exertion, brought on chiefly by the endeavour to have the exposition sparkle at every point with dialectical flashes of lightning. It is with dialectic's flashes of lightning, and with the elements in which they are prepared and out of which they develop, as it is with natural lightning: the lightning-fire must as a rule presuppose clouds, but the clouds are not on that account lightning.

rightly (Τὸν δὲ ἐρωτᾶν καὶ ἀποκρίνεσθαι ἐπιστάμενον ἄλλό τι σὺ καλεῖς ἢ διαλεκτικόν; *Krat.* 390 c.). That what essentially matters in dialectic is not the conversation but the development of concepts conditioned by “the art of language” goes without saying; but that the conversational form, when everything irrelevant is kept out, is nonetheless a very natural form for the beginning dialectician, may readily be shown by a few examples.

Something and Other. A. If the categories “something” and “other” simply excluded one another, there would be no dialectic; if they coincided without more ado, there would be no dialectic either. The dialectical shows itself precisely in this, that the two propositions “something is what it is, not an other” and “something is what it is not, hence an other” not only cancel but also mutually presuppose and ground one another. B. If the dialectical rests on so slippery a ground, then all dialectic is false. A. How so? B. The claim you have set up stands in open contradiction with sound human understanding, and *a philosophy that stands in open contradiction with ‘sound understanding’ is false.* A. If what you here call *open contradiction* is to be dismissed out of hand by an immediate verdict, philosophy can attach no weight to the ruling; for *a philosophy that conforms itself immediately to ‘sound understanding’ is superficial.* But if “sound understanding” will condescend to an inquiry, the inquiry can easily be got under way. B. It would have to be the second proposition, under whose negative form some ambiguity or other has hidden itself, and which might therefore become an object of inquiry; in the first proposition, the proposition of identity, the pure proposition of certainty (*principium certitudinis*), there cannot be a trace of ambiguity. A. But suppose it is after all the first proposition in which the ambiguity, the indeterminacy, and so far the uncertainty, lies hidden — what then? B. Then it must be capable of proof. A. Is it altogether forbidden in logic to say anything else of something than that it is something? B. By no means; for how, by merely repeating “something is something,” should one ever determine *what* something is? Logic, however, which has no fondness for ambiguities, draws a definite distinction, as easy to grasp as it is unmistakable, between determining something by means of an other and saying of something something other than what it is. A. So it is permitted to determine what something is by determining what it is not? B. Not only permitted but necessary. A. Then it is necessary that in the determination of something a determination of an other be contained. B. That too may be granted — but on the express condition that something, precisely by that determination, excludes the other. A. Something, then, is —? B. A thing different from an other. A. And if now, on that condition, we exclude the other, something is —? B. A different thing.[†] A. But now the

[†] *Translator’s note.* *Et Forskjelligt*, the substantivized neuter adjective: “a differing one,” something whose whole character is to be different. English has no natural form for it; “a different thing” is used

other, on its side, is —? B. A thing different from something. A. And when, likewise on that condition, we let the other exclude something, we say that the other is —? B. A different thing. A. So then: something, when determined, is a different thing; the other, when determined, is a different thing; something is therefore what the other is, namely a different thing. B. This ambiguity is easy enough to see through. Certainly something is a different thing, just as the other is; but different and different are two things — something is a different thing precisely by being different from the other, and the other a different thing by being different from something. Something and other are therefore still, as before, different from one another. A. Certainly something and other are different from one another; only a sophist would take it into his head to deny their difference. What was to be proved here was only this: that their difference is grounded in their identity. B. What is different is not identical, and what is identical is not different: only by pure categories can one secure oneself both against hidden ambiguities and against open contradictions.

A. But if I am not mistaken, you have already yourself, by your definite and lucid answers, made valuable contributions toward sharpening our eye for the dialectical unity of something and other. B. I should very much doubt it. A. But you did stress that something can be determined only as a different thing, and, for clarity's sake, as a thing different from an other? B. Yes, that I did stress. A. Then you have at the same stroke expressly required that something, in virtue of that difference, must be taken as an other than that from which it is different. B. Of course. A. So on your view something is other than the other, in opposition to the other — and hence, after all, an other. B. Your play on words leads nowhere; for when something is taken as an other, it becomes another other — namely the other that something itself is, as against all the other that something is not. A. That I can understand; now you are talking like a true dialectician. So you have now seen for yourself that, with respect to the determination contained in something, there are two kinds of other: an other that something itself is, as against another other that something is not — in other words, that something can exclude the other only by itself being an other. B. Yes — so long as one keeps to expressions as general as “something” and “other,” one can no doubt practise without much difficulty the game of thought of turning something into other and other into something. But let us try it with something determinate! Show me in definite examples that something really is identical with other, and I shall grant you that there is “sound understanding” in the dialectical. A. You take a leaf in a book to be something determinate? B. Naturally. A. And the two sides of a leaf

throughout the exchange, and the reader should hear the emphasis on *different* rather than on *thing*.

are surely so opposed to one another that if we call the front a something, we must call the back an other. B. Excellent! Now surely it will occur to no one to identify the front with the back, and then maintain that the front is what the back is, because both are sides, or because the front becomes the back when the leaf is turned, and the back the front. A. But on this occasion too the identity is surely as conspicuous as the difference? B. I can see no identity. A. What do you think, then, about the leaf? B. That the leaf which has the two sides consists precisely in the difference and opposition of the sides! For if the two sides were turned into one side, we should get only a surface, not a leaf. A. And if the two sides were separated in such a way that each came to belong to a body of its own, we should get two surfaces but no leaf. B. Yet it makes an essential difference whether I say "in the leaf the front is united with the back" or "the front is what the back is, the front is back." The first I can grant, the second not. A. If the leaf that unites the two sides were a third thing subsisting on its own over against the sides, we should certainly not be entitled to take it as the existing unity of the sides; but since the leaf, taken in separation from the sides, does not exist at all, the one side must in the leaf show itself identical with the other. B. Are we then perhaps to derive from this that the front itself is both front and back, and the back in turn both back and front? A. That is a consequence you will hardly reject. If we call the two sides α and β , it is evident that α , which in and for itself is neither front nor back, must by its opposition to β be front in so far as β is back, and conversely; α is therefore, in virtue of its identity with β , both front and back. B. That is a peculiar extension of the concept of identity. What I have to object against the concept of identity so extended I would rather hold back, however, until you have produced one more example of a different sort.

A. What do you say to calling the greater a something and the lesser an other? B. That example I find very suitable. A. And what is it, then, that we are to prove? B. What else than that something is both a greater and a lesser, the other both a lesser and a greater, and one and the same magnitude itself both lesser and greater. And in truth nothing is easier than to prove all this. A cursory comparison of three magnitudes, $\alpha > \beta > \gamma$, tells us at once that α is a greater in so far as β is a lesser, and β a greater in so far as γ is a lesser. But what follows from this? That β is in one respect a lesser and in another respect a greater — in that there is no contradiction; but if anyone maintains that the magnitude β is at one and the same time and in one and the same respect both greater and lesser than itself, he does fall into contradiction, and not only with the natural understanding but with true logic. That one and the same side of the same leaf can at one instant be front and at another back, and at one and the same instant — when, mark you, it is regarded

from two opposite points of view — both front and back, contains no contradiction, but a confirmation of the old rule *diversi respectus tollunt omnem contradictionem*. If, on the other hand, one ignores the *diversi respectus*, overlooking what both Plato and Aristotle, for all the difference of their intellectual bent, so strictly inculcated — that in such inquiries what is asked about must each time be seen to be τὸ αὐτό, and regarded κατὰ τὸ αὐτό, πρὸς τὸ αὐτό, ἐν τῷ αὐτῷ χρόνῳ — then one may indeed dazzle the inexperienced for a moment with piquant turns of identity and surprising contradictions, but one will never convince the expert, never satisfy the sensible thinking that inquires after truth. A. Weighty words, and important! So now your “sound understanding” has almost got the better of contradiction and finished off dialectic; I say *almost*, for one small point still remains — a point we simply must have cleared up, and one you will surely be able to clear up easily: it concerns *diversi respectus*, and you have of course studied *diversi respectus*. B. Studied? It has never occurred to me that there was anything further to study in logic’s *diversi respectus*; I have always supposed it enough to remark in a general way that there are *diversi respectus*. A. Then it is surely necessary to distinguish between *respectus*, since there are, after all, *diversi respectus*. Of the two sides of the same leaf — to return to one of our examples — it undeniably holds in one respect that they are the same, namely that both are sides, and in another respect that they differ, namely that one is front and the other back; in one respect the same, that both are, say, written on, in another respect different, that one is, say, white and the other red, and so forth. Each of these different respects may in turn give occasion for new ones; for although it holds of both sides in one respect that both are written on, yet on this very point it also holds in another respect that they differ, namely that the one is written in black ink and the other in red. These doublings of respect can be continued indefinitely. If thinking is not to lose itself in an unsurveyable tangle of all manner of regards to all manner of respects and remarks and considerations and reflections, logic must necessarily distinguish not only between essential and inessential respects but also between respects that fall indifferently outside one another and respects that are contained in one another and therefore follow from one another with inner necessity. That one side of a leaf is written on does not of course entail that the other must be so too; one can take account of the former without taking account of the latter; the former can be without the latter’s therefore being, and no one but sophists would think of positing the latter as identical with the former. It stands otherwise, however, with being front and being back: here it is impossible to take account of the former without taking account of the latter, for these two modes of existence are contained in one and the same existent in such a way that the regard to

the former actually coincides with the regard to the latter — the former is, despite the difference, indeed in virtue of the difference, *identical* with the latter. True insight into the essence of identity, difference and contradiction depends on analysing the *diversi respectus* sufficiently.

B. But can any well-founded objection really be made against the proposition *A* is what *A* is? A. Among others, the objection that the proposition lacks a ground. B. But when the proposition is one with the principle — the principle of identity, the principle of truth, the principle of certainty, *principium certitudinis* — and is of course self-evident in every way, it needs no ground. A. Is it then immediately evident from the proposition *that A* actually is? B. No, not *that A* actually is, but *what A* is. A. Is it immediately evident *what A* actually is? B. Not exactly *what A* actually is, but that, *if A* is at all, *then A* is *what A* is. A. But suppose *A* is not at all? B. Then it is still what it is. A. So in order to be what it is, *A* does not even need to be? B. You understand very well what I mean. A. The question here is not of opining but of knowing; still — what is your opinion? B. That even if *A* is a non-being, it is still what it is, namely a non-being. A. But non-being is surely the same as what we otherwise call nothing, no thing? B. Naturally. A. So then, whether I say "no thing is what it is" or "every thing is what it is," it must at once be evident to sound understanding — by the principle of identity, of course, the principle of truth, the principle of certainty, *principium certitudinis* — that with both propositions I mean one and the same thing!

B. But can anything at all be adduced in defence of so self-contradictory a proposition as "something is what it is not"? A. Among other things this: that a proposition which by its open contradiction so boldly challenges thinking must awaken reflection and call for grounds, in that it demands a refutation. B. But how should one refute by grounds a proposition that immediately refutes itself? A. The proposition "something is what it is not" by no means refutes itself immediately. B. If what is unthinkable, absurd and senseless in this proposition is not self-evident, then what is self-evident, and what is sound human understanding? A. Sound human understanding by no means sees anything absolutely absurd in something's being able to be what it is not. B. So sound human understanding is supposed to be able to conceive that *something* — mark you, at a given time and in a quite definite sense — *is what it* — at the same time and in the same sense — *is not*; no, that really is too much. A. Yes, that really is too much; but so absurd a demand is made by the proposition *something is* — mark you, when its existence is grounded — *what it is not* only when the proposition is held one-sidedly in superficial, abstract immediacy. B. Well, that is another matter, if your defence rests

on an interpretation. A. And another matter too, presumably, if your refutation rests on an interpretation. B. Then prove to us, by means of your dialectical interpretations, that one and the same something can at one and the same time and in one and the same respect be both what it is and what it is not, and that one and the same magnitude can at once be both greater and lesser than itself. A. Such proofs belong to sophistic, not to dialectic. B. But to prove that something can in one respect be what in another respect it is not, or that one and the same magnitude can in one relation be a lesser and in another a greater, nothing is needed but *diversi respectus*, no dialectic at all. A. But to prove that something, though different from the other, is nevertheless essentially one with its other, because the being of both has the same source, the same root; to prove that the same magnitude is on one and the same ground both a greater and a lesser, since the two relations, the two respects, have one and the same point at which they meet, namely being a magnitude; to prove that the identity of something and other, of the greater and the lesser, contains precisely the ground of something's being in existence different from the other, and the greater different from the lesser — for that, dialectic is needed; for this dialectic can prove, but dialectic alone.

Concept and Existence. C. A dialectical development of the inner connection of the *a priori* concepts, and of their transition into one another, is at the present standpoint of the science no doubt to be acknowledged on the whole; but it is another question whether, because the dialectical plays a considerable part in philosophy, one is on that ground entitled not merely to reproach formal logic with not being dialectical, but even — on the strength of a Platonic expression which can always be taken in more than one sense — to transform the whole of philosophy into sheer dialectic. A. Had you followed the preceding inquiry attentively, you would, as to the first point, easily have convinced yourself that there was no thought here of delivering a detailed critique of logical formalism, but only of rebuffing the self-sufficiency with which one not seldom allows oneself, invoking “sound human understanding,” to consign every logical objection to the immediate validity of immediate first principles to the realm of fancy and sophistry; as to the second point, your misgivings could do with a clearer statement. C. *A priori* concepts — such as something and other, essence and semblance, ground and consequent, and the like — can easily be produced out of one another by the application of contradiction. Whoever posits the ground already implies the consequent; the ground is therefore, at bottom, both itself and the consequent.[†] But since the ground becomes a ground proper

[†] *Translator's note.* Danish *Grund* carries the whole weight here: it is at once the logical ground (*Grund* as against *Folge*, consequent), the “bottom” of a thing, and the “reason why.” Nielsen plays on all three, and the English can only follow at a distance.

only on account of the consequent, the ground is consequent and the consequent ground. The same thing is not merely ground in one respect and consequent in another; the two respects also coincide in a single respect: to dialecticize[†] the concepts out of one another is, if I may use the expression, the same as to dialecticize *diversi respectus* into one another. As for the *a posteriori* concepts, the concepts of experience, it is impossible to dialecticize existence out of the concept; or who would try to dialecticize the actual horse out of the concept of a horse, or the actual snail out of the concept of a snail? But if it is impossible to dialecticize the existent out of its concept, then presumably it is for the same reason impossible to dialecticize one concept of existence out of another — the concept of a horse, say, out of the concept of a fly, and this in turn out of the concept of a snail. In the world of experience *diversi respectus* fall outside one another, and where *diversi respectus* fall outside one another dialectic plays a very subordinate part. A. But if you could not succeed in dialecticizing existence out of the concept (cf. Propæd. p. 31), have you never tried going the opposite way — dialecticizing the concept out of its existence, as for instance dialecticizing your concepts of snails out of their corresponding snail-existence? C. One does not dialecticize concepts of experience out of existing objects; one *abstracts* them. A. So there is nothing one might call a dialectic of abstraction? C. Not so far as I know. A. But the concepts abstracted from the objects must surely be, not only in our consciousness, but also in the objects themselves? C. Naturally; for if the concepts were not in the objects, they could not be abstracted from the objects. A. But the object from which the concept is abstracted is surely not itself a concept? C. By no means. A. Yet the concepts, in so far as they are in the objects, may be said to exist? C. That the concepts of experience exist in the existing objects, no one can well doubt. A. So in one and the same existence there are two existents — the existing object, and the concept existing in the object? C. No, it cannot be taken that way; it is not the object itself but its material that is opposed to the concept. A. Then presumably we get three existents: the existing material, the existing concept, and that existence of the object in which both exist? C. No, one cannot take it that way either. A. How, then, is it to be taken? C. The material by itself is no separate existent, any more than the concept is; the material and the concept are two sides — two sides of existence — of the same existing one, the object. A. The existing one is, then, over against and in opposition to the two sides of existence, the existent in and for itself? C. No, one cannot take it that way either; for if we let abstraction remove the two sides of existence, the

[†] *Translator's note.* *Dialektisere*, Nielsen's own verb and deliberately ungainly in Danish too: to derive one thing out of another by dialectic. It is kept rather than paraphrased, since the whole dialogue turns on the question of what can and cannot be so derived.

existing unity will itself become an abstraction, a side of existence that does not exist. A. But how, then, is it to be taken? C. So that the existing unity is seen to exist in the two sides of existence, which precisely thereby come to exist in inseparable unity. A. Then however many parts, qualities and sides the existing object may otherwise have, we may now, on your account, assume that these parts, with their sides, qualities and mutual relations, are all equally material by virtue of the material and intellectual by virtue of the concept; every side of existence thus becomes an existential unity of two sides, the material and the intellectual. C. That we may quite safely assume. A. But what is material in the object is surely something sensuous,[†] which as sensuous cannot be thought, and what is intellectual in the object something non-sensuous, something thinkable that cannot be sensed? C. Perfectly correct. A. Then in the existing unity the unthinkable is after all a thinkable, and the non-sensuous a sensuous. C. That I do not understand! A. Then neither do you understand the existing unity; for how should the thinkable in an existing unity be able to penetrate the sensuous without the sensuous thereby becoming, by virtue of the thinkable, itself thinkable, and the thinkable, by virtue of the sensuous, sensuous? C. It almost looks as though there were something in what you say. A. The concept of existence, then, is a system not only of intellectual but also of material determinations — a system of intellectual sense-determinations, of sensuous intellectual determinations. C. Yes, so it is. A. But then the concept of existence betrays a most striking contradiction. C. What contradiction? A. That of demanding existence by setting itself against existence. C. What does that mean? A. If the whole system of sensuous intellectual determinations is posited in the intellectual form, then the sensuous vanishes as a moment of reflection sublated in the thinkable — that is to say, the existent transforms itself into the concept of existence, but existence goes out. If conversely the whole system of intellectual sense-determinations is posited in the sensuous form, answering to immediate sense-impressions, then the object steps into existence, but the concept of existence goes out. . . . C. No, the concept of existence does not go out; it is completed in existence: the existent is, by what we have said, nothing other than the concept of existence complete in every respect. A. But was it not you who, by what we have said, granted that the object from which the concept is abstracted was not itself the concept? C. Yes, but I was thinking then of the abstract and so far incomplete concept, not of the concept concrete and complete in itself. A. So it is after all possible to dialecticize existence out of the concept, once one has learned to dialecticize the concept out of existence? C. To prove that in deed would here amount to showing it

[†] *Translator's note.* *Sandselig* = *sinnlich*, pertaining to the senses; it carries none of the modern connotation of "sensual."

in definite examples.

A. Sound human understanding tells us that the horse is an animal, that the fly is an animal, and the snail likewise; but how do such judgments actually stand to “the principle of identity and of contradiction”? C. The matter is plain enough. A. Yes, the matter is that if we let formal identity hold for the predicate, we deny it in each subject; whereas if we let it hold for each subject, we deny it in the predicate. C. How so? A. If we posit unconditionally “animal is animal,” then the horse becomes a fly and the fly a snail; if we posit unconditionally “the horse is horse, the fly is fly, the snail is snail,” then animal (horse) is not animal (fly) — that is, the common predicate contradicts itself and must fall away. C. This difficulty can be overcome well enough. A. With dialectic or without? C. Without any dialectic at all! One need only bethink oneself of the factual characteristic marks of the concepts of existence, and of the relation of the disjunctive judgment to *diversi respectus*. A. That we must certainly hear! C. If we compare animals with plants, it is seen that all existing animal individuals have one thing in common: being animal. If on the other hand we compare the animal existences with one another, then one animal series shows itself, by given characteristic marks, different from another. If we compare the existences in the one series with the existences in the other, then the individuals within each series are alike; but if instead we compare the existences belonging to a given series with one another, particular marks will show how one class is distinguished from another, and so forth. The animal, then, in opposition to the plant, is what the animal is; but by general and constant marks of difference the animal, in opposition to itself, is either mollusc or articulate or vertebrate.[†] In opposition to articulates and vertebrates the mollusc is always mollusc; but in opposition to itself the mollusc is either a mollusc proper or a radiate. Now if one simply goes on in this way with the divisions (*divisiones* and *subdivisiones*) through every series, every class, every order, every family, guided by the marks established by experience, until by way of genera and species one has reached individuals — what contradiction is there in that, and what have such classifications to do with dialectic? A. Unless it be that dialectic finds a good deal to do with *diversi respectus*. C. There is surely no danger of that. A. Then tell us, on the strength of your completed classification: what actually is a horse? C. A horse (*Equus Caballus*) is, by the category of its kingdom, an animal; by the category of its series, a vertebrate; by the category of its class, a mammal; by the category of its order, a pachyderm; by its

[†] Translator’s note. *Bløddyr*, *Leddyr*, *Hvirveldyr*, and *Straaledyr* just below, are the Danish names of Cuvier’s four *embranchements*: Mollusca, Articulata, Vertebrata and Radiata. The classification was standard in 1861 and has since been abandoned; “radiate” covered what we should now distribute among the echinoderms, cnidarians and others.

family category, akin to asses and mules; and a species-category of its own within a genus-category of its own. A. A fine collection of categories, then, all of which had to combine here to yield the horse's concrete category of existence! If only one could now, with the help of *diversi respectus*, get it properly explained how these many different categories have actually grown together — not to say, been bundled together — into a single one! C. They of course enclose one another in such a way that the more general always encompasses the more special. A. Surely not, though, in the way one envelope encloses another, or one ring encircles another? C. No, they rest upon one another, the more special having in each case settled upon

the more general. A. The more general a category is, the more abstract it is; the more abstract it is, the further it is from existing: so the existent rests upon the non-existent? C. No, I meant to say: they rest upon one another in such a way that the more general is always borne by a more special, this by one more special still, until we come to the most special of all, which is one with the existent itself. A. So it is not being a horse that depends on being an animal, but being an animal that depends on being a horse? C. No, that is not how I conceived it either; indeed, at this moment I cannot, to speak plainly, say how I actually did conceive it.

II.

Philosophy and the Particular Sciences.

General Remarks.

“With respect to the a posteriori, the advance of knowledge will doubtless depend on new information’s being constantly gathered, new matters of fact tested, new discoveries made, and so on; but is the matter thereby settled? If the advance of knowledge in relation to the a posteriori is regarded from this side alone, it is easy to dismiss philosophy. If it cannot so much as exhaust the a priori — seeing that mathematics owes it no thanks — how much less can it meddle with positive matters of fact! Let the historian lay a problem before it: it cannot answer, for a historical problem is answered only historically. Let a chemist apply to it: it has no knack for experiment and makes no discoveries in chemistry. Let the astronomer want help with an observation: it scarcely understands him. Let the physician ask it for advice: it is ignorant. Even what it has speculated out on its own account concerning the relation of soul and body the physiologist may unfortunately be unable to use.” (Propædeut. pp. 99–103.)

The extent to which it holds that philosophy is dialectic is seen from its relation to the subject-sciences,[†] for that relation is dialectical in the strictest sense. If one overlooks the dialectical, the relation becomes ambiguous and doubtful; for while the subject-sciences are by nature inclined to dismiss philosophy, philosophy in its turn, when it conceives its vocation dogmatically, is tempted from its side to exalt itself at the expense of the other sciences. Schelling’s words are characteristic in this regard: “Speculation is everything;

[†] *Translator’s note.* *Fagvidenskab*, literally the science of a *Fag* — a subject, a trade, a department. “Subject-science” is used here, as in the translations of Nielsen’s other works, and its practitioner, the *Fagmand*, is “the specialist.” These are to be kept distinct from *de særskilte Videnskaber*, “the particular sciences,” of this part’s title.

it is to behold in God, to contemplate that which is. Science itself has worth only in so far as it is speculative, that is to say, contemplation of God as he is." (Propædeut. § 17, pp. 81–83).

What has chiefly contributed in recent times to confusing the issue is the circumstance that Hegel — whose penetrating grasp of contradiction, of the negative, and of the immanent method of development (Propæd. pp. 150–158) must place him so high in the ranks of the dialecticians — has nevertheless, in his account of philosophy's relation to the subject-sciences, overlooked the dialectic of the relation. Instead of acknowledging the scientific independence of the particular sciences, Hegel closes his propaedeutic survey of the encyclopaedia of the sciences (Propæd. pp. 84–85) with the declaration that the encyclopaedia is properly "an exposition of the general content of philosophy; for whatever in the sciences is founded upon reason depends upon philosophy, whereas whatever there is in them of what rests on arbitrary and external determinations, or, as it is called, is positive and statutory, lies outside it." But to let philosophy's superiority rest in this way directly upon the subjection of the subject-sciences is to take the matter undialectically; to see philosophy's independence grounded in its dependence upon the independent subject-sciences is, by contrast, to grasp the relation dialectically.

The independence of the subject-sciences over against philosophy is evident from the course of their development; for the methods and discoveries that mark the great advances of these sciences are so far from being attributable to philosophy that "the history of the inductive sciences" (Whewell) shows, on the contrary, to what degree independent research and a procedure answering to the nature of the case have depended on the researcher's being able to free himself from the one-sided influence of philosophical speculation (Propæd. pp. 167–68). What in the subject-sciences "is founded upon reason" by no means depends upon philosophy in particular. In the first place it is not reason that owes its origin to philosophy, but on the contrary philosophy that owes its origin to reason. And next — and in this connection this is decisive — it must be emphasized in the strongest terms, not only for the sake of the subject-sciences but above all for philosophy's own sake (Propæd. § 23), that there is a knowledge which is scientific in the strictest sense although it is not philosophical; indeed a thinking whose scientific energy depends precisely on its not being philosophical. For that in the empirical knowledge of the empirical there is a certainty of sensation, a many-sidedness of representation, an added gift of intuition which no philosophical glimpse of the Idea can replace, is surely as undeniable as it is undeniable that there is an exact and critical thinking not to be exchanged for the dialectical (Mathem. og Dialekt. p. 3). The rational distinctiveness of

the particular sciences is seen, too, from the fact that each one severally presupposes a scientific talent of its own answering to it.

But if one now sets up the proposition that *the subject-sciences are independent of philosophy, whereas philosophy — particularly as regards the appropriation of positive content — is dependent on the subject-sciences*, then this presupposition, true as it is in itself, may easily occasion a false inference. If philosophy, it is said, is so dependent on the exact and positive sciences in everything that concerns the exact, the empirical and the positive, then this dependence must plainly be a sign of its lack of independence. Every time philosophy comes into contact with some determinate positive subject matter, be its content mathematical, mechanical, chemical or historical, the philosopher must of course put up with being not merely set right about the facts but even duly schoolmastered by the specialist in the development of the concepts themselves. It looks, then, as though the concepts were perfectly trustworthy, clear, distinct and secure in all their consequences so long as they are preserved in the form in which the subject uses them, but abandoned to distortion as soon as they are made an object of philosophical reflection; it looks as though the philosopher had to borrow everything from the specialist, while the specialist for his part may regard the philosophical as a matter of complete indifference. This way of seeing things rests on a misunderstanding. Philosophy's dependence on the subject-sciences is, essentially regarded, a sign of its independence, of its ideal self-validity.

In contrast to the material objects of the surrounding world all sciences are of course ideal, all have ideality; but in contrast to philosophy, as the representative of ideality, the subject-sciences must be said to stand for reality. If the opposition is determined categorically, the real in its finitude will be recognized by its asserting itself through excluding its other, whereas the ideal proves its reality precisely by taking the other up into itself and relating itself to itself through the other. That mathematics as mathematics, chemistry as chemistry, natural history as natural history each keep to their own^{*)}, and, even when contributions are fetched from other subjects, still have first and directly to do each with its own, is precisely a mark of these sciences' finite reality. That philosophy, by contrast, must always have its ideal wealth in real poverty, its positive in its negative, was illustrated by Plato in an ingenious myth: at Agathon's banquet Plato has his Socrates, as is well known, present Eros, the ideal philosopher, as the son of Poros and Penia.

^{*)} "There is a cardinal rule which can never be violated with impunity, and which is nevertheless constantly violated in our groping advance toward the light. It is: *never to attempt the solution of the problems of one science by the order of ideas belonging to another*. There is one class of conceptions belonging to physics, another belonging to chemistry, a third belonging to physiology, a fourth belonging to psychology, a fifth belonging to social science. While all these sciences are closely connected, each has its own independent province, which must be respected." G. H. Lewes: *Hverdagslivets Physiologie*, Danish translation, part 1, p. 66.[†]

The realistic science, once founded, may in the course of time need manifold reshaping and undergo many changes, yet its supports and foundations will through it all remain unshaken. If the idealistic science, on the other hand, is to remodel its building, a *de omnibus dubitandum* will gradually shake every support; the whole building must then be pulled down to the ground, that it may be raised in its new style from the ground up. Each time such a critical juncture arrives it does indeed look for the moment as though reality were at stake, as though it were now all over with philosophy; but history confirms what also lies in the nature of the case, that such pervasive crises mark precisely the turning points in philosophy's rebirth.

"The subject-sciences are solid; they always have steady work, ample means and secure recognition in the republic of letters. Philosophy, by comparison, is like a traveller without settled provision, a hovering shape that sinks and rises, dies away and lives again, as the spirit will." (Propæd. p. 102). — Yet it is only from a wholly external, realistic standpoint that the ideal is taken as merely the vague, the indeterminate, the in itself indeterminable; the dialectical form is a form determinate and determinable in itself, and philosophy's relation to the subject-sciences is therefore to be stated categorically.

If philosophy is to be able to place itself in an inner and essential relation to a particular science, it must discover in that science a problem which it can also call its own problem. In order to show how such a *common problem* is chosen and treated, and so as not to scatter our attention over too many points, we shall in what follows confine ourselves to examples drawn from mathematics, chemistry and physiology; and we have thus, in mathematics, *the problem of infinity*; in chemistry, *the problem of change*; in physiology, *the problem of life*.

But even if such common problems are characteristic enough when viewed from the side of ideality, it is by no means thereby settled that they show themselves so when they are grasped realistically. Because the philosopher has noted some pervasive problem in one or another determinate subject-science, it is not yet said that the specialist will acknowledge the same problem in a corresponding way. A chemist who wished absolutely to resist a philosophical conception might, for example, reason thus: "There is in chemistry properly nothing at all that one may call a problem of change, but rather a concept of change, which arises and comes to be applied in so far as chemistry investigates the natural laws according to which the inner changes of bodies take place. But it is one thing to investigate the laws according to which the changes take place, and another to want to fathom the essence of change, to speculate about its possibility and impossibility — a task wholly foreign to chemistry." In the same spirit a physiologist

wishing to dismiss philosophy might reason thus: "Physiology does indeed proceed from the presupposition that there is a something we must call life, since in living organisms there is a law-governed system of vital phenomena; but since physiology essentially confines itself to investigating the phenomena, without penetrating into what lies hidden behind them, it can by no means be granted that there is here any proper problem of life. What life is in itself — a principle, a force, a product of the peculiar cooperation of mechanical and chemical forces — no matter! Let each form whatever individual representation of the essence of life he pleases; so long as he is not thereby kept from seeing the validity of the laws and demonstrating the physical causes, physiology is satisfied."

The difficulties which the natural influence of differing standpoints on the conception of a scientific problem always brings with it are considerably heightened by the ease with which something human — that is to say, something one-sided, accidental, arbitrary, passionate — mixes itself into the investigations. And since philosophy, for all its ideality, is like every other science subject to human conditions, it becomes important in this connection to take the matter from the purely human side as well, and so to note the obligations and rights which the sciences have to uphold in their mutual conflicts.

That the common problems in question must look otherwise in philosophy than in the subject-science is a matter of course. The problem of change, even if the chemist will acknowledge it in his own way, is in chemistry by no means what it is in philosophy; likewise the problem of life, even when the physiologist grants its reality, is one thing in physiology and another in psychology. The philosopher who knows the problem in the form of otherness, and is aware that the same is nevertheless not the same, must therefore make it his chief business to clarify the conditions of the development and to make the transformation plain. Anyone who would seek guidance in a particular science must of course make himself at home in its way of thinking. If the philosopher neglects this — if he misunderstands the character of the science in question either as a whole or in essential particulars, whether by ascribing to it an aim it does not have and cannot have, or by tearing its doctrines out of their proper connection and so distorting them, or by overlooking the peculiar presuppositions and conditions under which a given concept with its corresponding designation was formed, thereby laying a false meaning under the concept and in this way coming to force upon the science views, results and consequences it cannot acknowledge — then the specialist is within his rights in speaking out. This right becomes a formal duty to the truth when such blunders on philosophy's part, as has not seldom been the case (*Propæd.* pp. 170–171), are combined

into the bargain with the conceit that it is philosophy which from its exalted standpoint must feel called upon to reform the subject-sciences. But further than a rejection of philosophical encroachments the specialist cannot and should not go. If he goes further — so that, for example, he is not content to secure to the concept the meaning it has in his science, but regards it as belonging to his subject to such a degree that he either flatly forbids the philosopher to touch it, or at any rate takes it upon himself to prescribe how far, in what manner and in what connections philosophy may develop the borrowed concept graciously made over to it by the subject-science — then there is to be seen in this not merely a presumption but a ludicrous misunderstanding.

"Confined to the several compartments of consciousness, the concepts gained are still so grown together with finite constituents, still so weighed down with particular presuppositions, that they find themselves as it were in a state of unfreedom. They are held here as the prisoners of their matter within a determinate horizon; they are, so to speak, the subject's bondsmen, who dare not leave the appointed place of work." (Propædeutik p. 113.)

Now in setting out to free the real concepts from what is accidental in their delimitation, to unfold the consequences and to bring the schematic forms into dialectical flow, philosophy must of course — once it has drawn the concepts into its own horizon, posed the problems in its own way, and thus taken the whole responsibility of the treatment upon itself — have an indisputable right within its own province to apply its own measure, to introduce its own designations and usage, and to carry out the development in its own spirit. If it stands firm that every subject-science must in essentials contain its own correctives, then philosophy too must surely contain its correctives. A judgment upon the errors and blunders a philosopher may possibly have committed in carrying through the dialectical analyses belongs in the last instance not before the forum of the subject-science but before that of philosophy.

To deny philosophy the full scientific right to treat in a free philosophical manner what it has appropriated from the subject-sciences is to deny it a development of concepts of its own, and to resort to a dictum of authority. If it is to go by dicta of authority, then a specialist may settle philosophy's fate as easily as that caliph settled the fate of the Alexandrian library: "Either it contains only what stands in the Koran, and then it is superfluous, or it contains something else, and then it is false"; — "either philosophy contains only what stands in the subject-science, and then it is superfluous, or it contains something else, and then it is false." It is a Saracen dilemma; but a Saracen dilemma cannot make philosophy give up its right.

A.

Philosophy's Relation to Mathematics:**The Problem of Infinity.**

"A philosophical fox or marten has stolen in upon mathematics' fat poultry to fill his schematic belly; he means to suck out their warm blood, so that nothing is left but the dead carcasses, void of sap and strength, which are then to be presented as mathematics' vigorous, practical, lively concepts" (Professor Steen's "Afregning," p. 7).[†]

These boding — indeed ill-boding — words may well seem at first glance to have something particularly forbidding about them, especially when set beside a declaration such as this: *"In the first place there is nothing that one may name 'the problem of infinity' in mathematics"* (cf. Begrebet: Nøiagtighed, p. 38). But when it is thereupon conceded that there is *"a mathematical concept 'infinite' which arises and comes to be applied, and which is extremely important and moreover difficult, because it is of a mixed nature"*, the forbidding element disappears and something inviting shows itself. For if the mathematical concept "infinite" which arises and comes to be applied is, on the mathematician's own admission, *"extremely important and moreover difficult, because it is of a mixed nature"*, then this contains, so far as philosophy is concerned, a well-founded invitation to investigate whether this concept of infinity, in mathematics so *extremely important and moreover difficult*, may not after all be connected in one way or another — perhaps without the mathematician's knowing it — with a problem running through the whole of mathematics; whether philosophy may not have every reason to acknowledge, in this pervasive problem, its own age-old problem of infinity; and whether mathematics, without a mathematician's always quite thinking of it, may not be said to furnish essential and indispensable contributions toward that problem's solution.

a) The Concept of Infinity.

That the concept "infinite" which in mathematics "arises and comes to be applied" is "extremely important" we may grant at once; but we must most definitely deny that the

[†] *Translator's note.* This whole section is a reply in an ongoing quarrel. In 1860 the mathematician Adolph Steen attacked Nielsen's *Philosophie og Matematik* in a pamphlet called *Afregning* ("Settling of Accounts"), and Nielsen answered with two pieces of his own — *Hr. Professor Steens "Afregning": Hermed en specificeret Opgjørelse til Revision* and *Om Begrebet Nøiagtighed med særligt Hensyn paa Hr. Prof. Steens "Bidrag"*, the latter a postscript to *Philosophie og Matematik*. The short titles Nielsen cites here and below — *Afregning*, *Begrebet: Nøiagtighed*, *Om Begr. Nøiagt.* — are to those pieces, and the words he puts in quotation marks are his opponent's objections as he reports them there. Without this the pages read as an argument with nobody.

concept of infinity, taken as a concept by itself, is either difficult or of a mixed nature. Of the infinite in general only this can be said: that it is a negation of the finite, the not-finite. If any difficulty lay hidden in such a negation, the difficulty could not stem immediately from the negation itself, but would have to be referred essentially to what the negation cancels. So if one is to find difficulties in the mathematical “infinite,” one must find corresponding difficulties in the mathematical “finite”; and if one raises no objection to the latter, it betrays mere thoughtlessness to raise objections to the former.[†] If the concept of infinity is difficult “because it is of a mixed nature,” would not the concept of finitude, when the finite is specially emphasized, be of a still more mixed nature? If it be objected that the mathematical “finite,” which holds only of magnitudes and not of things, cannot possibly be of a mixed nature, then one will also be obliged to grant that the negations in which such a “finite” disappears can as little be so.

Concepts of a mixed nature always have a questionable admixture of the conceptless, and as mixture-concepts they stand on the lowest rung in the order of concepts. How does a mixture-concept come about? By parts of unlike kind, brought together in an external and accidental way, being subsumed under a collective unity, which in consequence comes to contain marks of unlike kind accidentally brought together. Thus the concepts “topsoil,” “gravel,” “marl” are genuine mixture-concepts; but a concept of gravel and a mathematical concept of infinity — what a difference!

But, someone may perhaps ask, if the concept “infinite,” in however many and various connections it occurs and however many transformations it undergoes, always preserves one and the same unalterable fundamental meaning — namely that of being “the negation of finitude” — how does it come about that so pure, so negative a logical concept can in a mathematician’s eyes take on the appearance of being “of a mixed nature,” or, what is the same, of belonging among the mixture-concepts? The answer is not difficult: *one mixes the accidental with the essential; by this mixing one turns up a number of meanings of differing kinds; one does not derive these separate meanings from the fundamental meaning, does not exhibit the necessary transition of the one into the other, but sets them up externally side by side, makes them an object of enumeration — and behold, the thing goes as of itself.* If one puts such meanings as “negation in positive form,” “impossible,” “transition involving something qualitative,” and so on, externally together, one gets at once an impression of the mixed. How is one to distinguish a negation in positive form from a negation in negative form, except by the former’s expressing the negative mediately

[†] *Translator’s note.* Danish *gjøre Ophævelser over* is an idiom, “to make a fuss about”; but *Ophævelse* is also the technical word for sublation, and Nielsen has just used its verb (*ophævet*, “cancelled”) two sentences earlier. The pun does not survive.

and the latter immediately? But if now the finite is set against the infinite, it is not the infinite that expresses the negation in positive form but on the contrary the finite, in so far as the finite, though it has a positive form, is yet mediately determined as the not-infinite. To take negation's own immediately negative expression for its positive form betrays a comical confusion. If someone, to display his acumen in matters logical, were to maintain that the proposition " a is not something" was straightforwardly negative, whereas the proposition " a is nothing" was affirmative, the negation being lodged in the word "nothing," one would surely agree without difficulty to assign such trials of acumen — formulated in all clumsiness after a formal logic's S is $Non - P$ — if not outright to the pedantic, then at any rate to that class of immature attempts which betrays that the acute man has not yet learned to distinguish between fruitful and unfruitful distinctions. If one now grants, so as not to be saddled with too much logical mixed goods, that the proposition " a is nothing" is just as directly negative as " a is not something," then one will also grant that the proposition " a is zero" is just as directly negative as " a is nothing." But if it stands firm that the sign 0 is just as direct and immediate an expression of negation as the word "nothing," then it takes a quite abnormal acumen to see that, while the proposition "between a and a there is no difference" is straightforwardly negative, the equation $a - a = 0$ should on the contrary be so nailed fast in the positive that only the screw of a paraphrase could bore through to the negation. (Cf. Om Begrebet: Nøiagtighed, p. 39).

The logical construing of mathematical equations is a peculiar business; the more weight is laid here on the conceptual side, the more evident it becomes that philosophy has a word to say here too. On the natural mathematical way of thinking, the equation $\frac{a}{0} = \infty$ shows us the well-known relation between divisor and quotient in its extreme consequence: if, on the assumption of a constant dividend, the divisor goes on shrinking to infinity, the quotient must go on growing to infinity. There is nothing in this relation to recall the impossible; everything points on the contrary to the possible: to the smallest possible divisor there answers the greatest possible quotient. If, on the other hand, in order that the infinite may come to signify the impossible, one chooses another point of view and construes the equation thus — "When it is asked how many times 0 must be made greater in order to give a , the answer in words is: that cannot be produced by multiplying 0; but in signs: $\frac{a}{0} = \infty$, 0 must be made infinitely many times greater in order to give a " (Om Begr. Nøiagt. p. 40) — then such a construal is well fitted to promote all manner of admixtures of an illogical miscellany. If the equation $\frac{a}{0} = \infty$ were really and seriously designed to answer the in itself absurd question "how many times must

0, the absolute 0, pure nothing, be made greater in order to give a ?”, then notations such as $\frac{a}{0} = 0$, $\frac{a}{0} = a$ would have been far more adequate. For from such expressions the impossible, the absurdity in the demand, is seen instantly, whereas the equation $\frac{a}{0} = \infty$, by its ironic pointing to the infinite, only mimically hints at what is absurd in the question, and then — quite rightly — transfixes the questioner by “a negation in positive form,” because he is asking not as a mathematical thinker but as a logical fool. If one gives up the logical foolery of trying to reckon how many times pure zero must be made greater in order to produce a ; if, in order that the equation may yield any sense, one lets 0 denote the smallest possible divisor while ∞ denotes the corresponding greatest possible quotient — then it is seen from every side how the impossible gives way to the possible. From the equation $\frac{a}{0} = \infty$ there follows directly, in virtue of the concept of the relation in question, $a = \infty \cdot \frac{a}{\infty} = \infty \cdot 0$; likewise from $\frac{b}{0} = \infty$, $\frac{c}{0} = \infty$, $\frac{d}{0} = \infty \dots$ there follow $b = \infty \cdot \frac{b}{\infty} = \infty \cdot 0$, $c = \infty \cdot \frac{c}{\infty} = \infty \cdot 0$, $d = \infty \cdot \frac{d}{\infty} = \infty \cdot 0 \dots$ Instead of the impossibility of $\infty \cdot 0$ ’s being able to give a , we see on the contrary the possibility of $\infty \cdot 0$ ’s giving not only a but b , c , $d \dots$ — in short, as many different magnitudes as you please. Instead of the curt, dismissive “impossible,” we get possibility upon possibility, possibilities in abundance. This result is of course so far from surprising the *thinking* mathematician that it does not even surprise ordinary sound human understanding. Everyone understands that any finite magnitude whatever may — and this is a possibility — be said to consist of infinitely many infinitely small parts, hence $a = \infty \cdot \frac{a}{\infty} = \infty \cdot 0$, and that it may therefore also — and this is a possibility — be divided into infinitely many infinitely small parts, hence $\frac{a}{\infty} = 0$.

The fixed idea that the concept “infinite,” as a “negation in positive form,” should be better fitted to denote “the impossible” than the concept “finite” is easy to clear up. For “the impossible” is in and for itself as indifferent to “infinite” as to “finite”; the impossible shows itself every time a demand that has been set up annihilates itself, and such self-annihilating demands may just as well presuppose the finite as the infinite. It is, to stay with the infinite, impossible to *enumerate* all the terms of an infinite series; it is, to pass to the finite,

impossible for a human being to *count up to* a quadrillion; and it is — here we must lean at once on both the finite and the infinite — impossible that a sphere with an infinite radius should be a sphere, or a plane a spherical surface with a finite radius; and just as impossible that the equation $a = 2a = \infty \cdot a = \frac{0}{a}$ should have any reality. If in the language of mathematical signs one wants a more direct notation for something impossible in itself, then it is not 0 and ∞ but $\sqrt{-1}$, $\sqrt[n]{-a}$ and the like that one has to keep to. (The

“Afregning” as revised by R. N. Appendix, pp. 26–27).

In logical generality it is easily seen that the concept “infinite” does not as a concept take up constituents of unlike kind, and so cannot be reckoned among the mixture-concepts. That the concept nevertheless occurs in several meanings, shows itself in a different light, and to that extent may, when treated superficially, acquire a tinge of being “of a mixed nature,” has its ground in something quite different: to think the concept through is to unfold the problem.

b) The Problem of Infinity. Suppose someone were to open an objection to a philosophical reference to the way in which the problem of finitude finds its solution in mathematics with the following declaration: “In the first place there is nothing that one may name the problem of finitude in mathematics, but rather a mathematical concept ‘finite’ which arises and comes to be applied, and which is extremely important and moreover difficult, because it is of a mixed nature.” One would surely find that a strange thing to say. What a finite magnitude is, is soon said; but the problems concerning the mutual relations of finite magnitudes are countless. Instead of saying that there is no problem of finitude in mathematics, one would rather have to say that there are countless problems of finitude in mathematics. If these problems fell apart from one another in loose multiplicity, one might indeed deny that there was any general problem running through them all; but one would then at the same stroke have to deny that there was any mathematical method — in other words, that mathematics was a science. If mathematics is a science, if there is a method by which the various problems of finitude are framed and solved in systematic connection, then there is also a general problem running through them all, a fundamental problem embracing and determining the method in all its ramifications, a problem whose all-round solution demands nothing less than the whole of “the analysis of the finite.”

But what holds so incontestably of “the problem of finitude” in its relation to “the analysis of the finite” — should this not hold also of the relation between “the problem of infinity” and “the analysis of the infinite”? What holds so incontestably of the concept “finite” in its relation to “the problem of finitude” — namely that the difficulty lies not in the concept but in the problem — should this not, on closer inspection, be found to hold also of the concept “infinite” in its relation to “the problem of infinity”? —

α) Concept and Problem. As the distinguishing mark of mathematical concepts — however much they may otherwise resemble the general logical ones — it must be expressly emphasized (Mathem. og Dial. pp. 21–27) that they are operational concepts. From this it follows that, while in their generality they present no difficulty, analysing

thought will soon discover that their proper development must always take place in the closest connection with the operations whose logical essence they are meant to clarify. What a logarithm is can easily be stated in a definition exact enough for sound understanding to hold the concept fast; but the manifold and characteristic consequences in which the concept's content first really shows itself come forth only in the course of the operations. What is meant by dividing a line in extreme and mean ratio — or, as it is also called, by golden-sectioning the line — is soon said and easily grasped; the concept "golden-section"[†] presents no difficulty. The difficulty, if there is one, is to solve the problem of the golden section and to develop all the consequences contained in it. What holds of the particular operational concepts holds of the general ones too: only from the systematic solution of the problems of finitude does it come to light what mathematically lies in the concept "finite"; only a carrying-through of "the analysis of the infinite" can open us a scientific insight into the mathematical concept of infinity. From the consequences of the operations it is further seen that the infinite falls not only outside but also inside the finite (cf. *Phil. og Mathem.* pp. 20–24; *Mathem. og Dialekt.* pp. 28–38), and that it develops in manifold ways out of finitude itself. Had a pervasive problem of infinity not originally stirred beneath "the analysis of the finite" and declared itself in many separate flashes, the differential and integral calculus would never have been developed. When Leibniz solved "the problem of tangents," what was essentially at stake was not the tangent and the secant (cf. *Phil. og Mathem.* p. 64) but the articulating reciprocity of finite and infinite: in "the problem of tangents" "the problem of infinity" is to be seen.

β) *The Mathematical Conception.* That thinking mathematicians, however much weight they may otherwise lay on the satisfactory results of the operational method, have nevertheless never been able to separate satisfactorily the concept they find inconceivable from the method that rests upon that concept, is shown by the history of mathematics from Leibniz and Newton (cf. *Mathem. og Dial.* pp. 90–91, p. 95 and elsewhere) down to the present time. "In the first place, when the principle is unknown we cannot know whether all the consequences following from it can be admitted: we go as if blindfold, and it might well happen that after laborious effort we should at last strike upon something which exhibited precisely the nature of the principle — that is to say, upon something we could not conceive. In the second place: is it not a shameful degradation of reason to make it follow a concatenation of terms, a series whose beginning and end

[†] *Translator's note.* The Danish verb is *høidele*, literally "to high-divide," the nineteenth-century Danish term for division in extreme and mean ratio, i.e. the golden section. English has no corresponding verb, so "golden-section" does duty as one here and in the next sentence.

it cannot see through? It is a great evil that is born of that kind of routine: accustomed to operating without seeing the ground, the mind loses all practice in thinking, and ends by carrying everything out quite mechanically.” (Mathem. og Dial. p. III). These misgivings of Savrien’s sufficiently explain the unrest and tension in which the analysis of the infinite has constantly held mathematical reflection^{*)} (Mathem. og Dial. pp. 55–56), whose problem — as a prize question set by the Berlin Academy in the year 1784 characteristically puts it, “wie aus einer widersprechenden Voraussetzung so viele wahren Sätze haben können hergeleitet werden” [how so many true propositions could have been derived from a contradictory presupposition] — will not cease to return, even though “routine,” the mechanical facility, will at all times no doubt afford “*habiles Calculateurs*” satisfaction so ample that it may easily escape this or that reckoning-master’s notice that there is in mathematics something one may name “the problem of infinity.”

γ) *The Philosophical Conception.* “To make the problem of infinity recognizable, reflection must lean upon the operation; to solve the problem, it must appropriate the intelligence expressed in the operation. That there can be a finite ratio between infinite magnitudes; that a sum of infinitely many terms may in some cases make up a finite and in others an infinite magnitude; that a series whose terms are so related that the limit of the ratio of two successive terms gives the quotient 1 may in some cases be convergent and in others divergent, and so on — such things will everywhere escape the thinking that does not, at the operation’s hint, catch sight of the exact determinations. That these determinations must in philosophical form become dialectical, while in mathematical form they neither are nor can be so, shows a characteristic difference between mathematics and philosophy, but by no means any conflict. No concept, however exactly it be defined, can maintain itself in a being indifferent to the other concepts: the validity of concepts is seen in their transitions; but the transitions which exact analysis makes by the operation, the thinking that penetrates the operational concepts makes by dialectic” (Mathem. og Dial. p. 57. Cf. pp. 59–60). If the finite is set immediately against the infinite, sound understanding involuntarily gets the impression that it is the finite which, in its multiplicity of terms and relational determinations, contains the wealth of existence, whereas the infinite is a bottomless nothing, waste and empty as empty space. And since thought can determine only what is in itself determinable, and only that is determinable which admits in some way of being defined, and only that admits of being defined which admits of being bounded, and the bounded is a finite — it must

^{*)}See Carnot: *Reflexions sur la Méthaphysique du Calcul Infinitésimal*. Paris 1797 (übersetzt von Hauf. Frankf. a. M. 1800). Versuch einer Philosophie der Mathematik — mit einer Kritik der Aufstellungen Hegel’s über den Zweck und die Natur der höheren Analysis — von H. Schwartz. Halle 1853.

constantly appear to the understanding as though only the finite were conceivable, while the infinite, the unbounded, were really not to be conceived at all, but, like the ancients' hyle (τὸ ἄπειρον), to be known by a kind of spurious thinking (λογισμὸς νόθος).[†] Where this misunderstanding has taken hold there can of course be no question of entering more closely into the problem of infinity; to see the problem's philosophical significance one must absolutely have at least an inkling of the dialectical reciprocity of the finite and the infinite. "An angle of 90^0 can be divided into infinitely many infinitely small angles, and so can an angle of 1^0 ; a square mile can be divided into infinitely many infinitely small squares, and so can a square inch: what follows from this? If the emphasis is laid on the resolution of the finite into the infinite, the consequence does indeed seem to be that there is no longer any difference between the angle of 90^0 and the angle of 1^0 , that the part becomes equal to the whole — that the boundary, in other words, is effaced. But if the emphasis is laid instead on the infinite's existing in and through the finite, then there is not only a difference between finite and finite, between the part and the whole, but at the same stroke a definite difference shows itself between infinite and infinite, with respect to the infinitely small as well as to the infinitely great. The one infinite number of infinitely small angles may then be as different from the other as 1^0 from 90^0 ; the one infinite number of infinitely small squares as different from the other as a square inch from a square mile. That the boundary is sublated in the finite means that it is posited in the infinite; that the infinite takes the boundary up into itself means that the finite is posited. The dialectical unity shows itself in this, that the boundary belongs as much to the infinite as to the finite" (Mathm. og Dial. p. 33). The fundamental thought that the infinite not only effaces the boundaries of finitude but also takes the sublated boundary-determinations up as articulating moments in its own development of concepts is, from the standpoint of the operations, sufficiently designated by saying that the infinite is analysed.

b). The Analysis of the Infinite. In the task of "analysing the infinite," then, mathematics and philosophy have a comprehensive common problem. That the problem must, however, undergo a transformation in being taken up by philosophy, and that the philosophical analysis neither can nor should be carried through in the same way as the mathematical, is a matter of course: for it is one thing to develop a concept for use in the operations that fall under it, and another to reflect the operations through in order to make plain the concept at work in them. "Where several sciences treat the same

[†] *Translator's note.* The phrase is Plato's, *Timaeus* 52b, where the receptacle of becoming is said to be apprehended by a "bastard reasoning" — λογισμῷ τινι νόθῳ — and hardly to be believed in. Nielsen pairs it with *hyle*, matter, and with the Aristotelian τὸ ἄπειρον, the unlimited.

thing, each will make its own clear; but they will never make the same thing clear in the same way. What one development must by its aim determine with the greatest exactness, another may by its aim only hint at in fleeting, hovering strokes, precisely because it has to let the emphasis of exactness fall on other points.” (Begreb. Nøiagt. pp. 51–52).

Since mathematics as a subject-science looks directly to its own, it cannot as a rule be presupposed that mathematicians as specialists should have either any particular interest in, or any special facility for, following philosophy’s dialecticizing analyses. But it may be presupposed — indeed, in the name of scientific standards, plainly demanded — that the specialist who feels called upon to pass judgment on what is distinctive in a philosophical procedure should at least grant the matter enough attention to hold fast the most essential starting points, follow the train of thought, and know what is being talked about. When, to cite a couple of examples, it is expressly to be brought out what difference it makes whether the categories opposed to one another are *division* and *composition* or *piecing-together* and *motion*, and this is then illustrated by a comparison between the broken and the curved line thus: “Of the broken line a composition holds, for its parts are finite; it is a piece-work belonging in each and all to finitude. Of the curved line no finite composition whatever holds, no piecing-together, not even the very finest; it owes its coming-to-be to motion alone, for it belongs to infinity” (Phil. og Mathm. p. 76) — would not anyone who grasps what is being talked about see at once the impossibility of characterizing the category of *piecing-together* without presupposing the immediately given existence of the finite parts? If we are to reflect here on how all the lines out of which the broken line is composed — the parts out of which a whole comes to be — themselves in turn have their becoming, then we are looking at becoming alone, not at piecing-together. Here too, of course, in so far as it is not a matter of clarifying a thought but only of lodging “a critique,” one may on the subject’s behalf lodge the following: “the straight and the broken line owe their coming-to-be to motion neither less nor more than the curved line does; the curved line is mathematically composed of finite lengths, just as the broken and the straight are.” But what does science gain by such a critique? To let everything run together — formation by piecing-together and coming-to-be by smooth transition — to let everything come to the same thing: discreteness and continuity, finite and infinite, broken and unbroken, curved and straight, straight and crooked — that is surely as little mathematical as it is philosophical!

Ignorance and thoughtlessness are by no means always the same: one can be ignorant without being thoughtless, and one can be knowing in a finite sense and yet thoughtless to a striking degree. The examples are manifold. In a philosophical investigation the

question is raised how the line's origin is really to be conceived in relation to the point; for that one always has the line once one has the surface, and the surface once one has the body, goes without saying. "In relation to the line as the positive, the point, the line's boundary, is the negative; when the line is a representation, the point is a thought. In relation to the surface as the positive, the line, the surface's boundary, is the negative. The surface is represented when the line is thought." (Phil. og Math. pp. 65–67). When, then, the question is how the line's origin is really to be thought — since letting the line arise by "composition of finite lengths" is merely to presuppose what was first to be shown, the finite lengths being themselves lines, and since letting the line's coming-to-be rest on surfaces will not do either, the surface's own origin necessarily presupposing the line — when this is expressly the question, and a man of the subject then, an expert who knows all the cases, with lines and with surfaces, with intersection and without, comes forward with the information that "*mathematically, curved lines also arise from the mutual intersection of surfaces*": then one receives not merely a piece of mathematical information but an aesthetic encouragement into the bargain.

α) Boundaries. The difference between exact and dialectical thinking is recognizable above all in a different use of the boundary. "Either something or nothing," "either positive or negative," "either finite or infinite," "either attainable or unattainable": such judgments of separation presuppose a sharp stress on what divides in the boundary. The sharply dividing boundary — the boundary that holds what it separates so entirely outside one another that the one must be denied when the other is affirmed — is *qualitative*. But the more the contradictory opposition is sharpened, the more exclusively the qualitative holds, the more conspicuous it becomes that the boundary, by being what divides, is precisely what unites. If one posits "either $+a$ or $-a$," the positive is indeed thereby held outside the negative; but since the division is so sharp that there is nothing whatever between, the positive and the negative are brought into immediate contact precisely in the dividing boundary, the zero point. The zero point is therefore the point of union of the positive and the negative just because it is their point of separation. If we now put the finite for the positive and the infinite for the negative — "either finite or infinite" — the same thing repeats itself. By staring at the negation, at what divides in the boundary, one gets the impression that the finite and the infinite are held apart by a yawning abyss: "either finite or infinite! One may therefore go on with the finite as long as one likes; a finite is and remains a finite, never an infinite." But by bethinking oneself of what unites in the boundary one easily sees that things so divided must nevertheless lie infinitely near one another: "either finite or infinite! So when one does not have the finite, one

has straightway and without more ado the infinite.” Where the finite ceases the infinite begins; the boundary, the negation, is therefore itself the logical point of contact, the point at which what is divided meets: here there is nothing between. “But if the finite and the infinite are in this way each other’s boundaries, the determination of boundary must become dialectical. For if the infinite has its boundary in the finite, then it has a boundary of finitude, a finite boundary, and is itself finite. If conversely the finite first reaches its boundary in the infinite, then it has no finite boundary and is consequently infinite.” (Phil. og Math. p. 22). The infinite is an unbounded; how, then, can that which has no boundaries at all be itself a boundary? The finite, which everywhere has boundaries in so far as one finite bounds another, may rightly be said to be something bounded in itself; but how can what is bounded in itself have an unbounded for its boundary? Or are there perhaps three kinds of boundary here: a boundary that holds for the finite and not for the infinite, a boundary that holds for the infinite and not for the finite, and a boundary common to finite and infinite? But if this is assumed, how does each of these boundaries stand to that whose boundary it is?

What, in respect of the mathematical mode of expression, easily gives the concept of boundary an ambiguous tinge is the circumstance that both magnitudes and negations of magnitude are posited as boundaries, and that 0 and ∞ cancel one another. By 0 the finite boundary is posited; by ∞ every finite boundary is annulled. Closer consideration will then easily make it evident that both negations belong to a complete grasp of the quantitative boundary. Without quality no quantity, without a zero-boundary no magnitude; but without a quantifying which in turn annuls the immediate zero-boundary and transforms the qualitative into a moment vanishing under increase and decrease, there is no magnitude either^{*)}. To see this one need only set 0, 1 and ∞ side by side; for while 0 denotes the posited boundary, the abstractly qualitative, and ∞ the annulling of all possible boundaries, pure quantifying, the numerical unit itself is conditioned by both. If we say that a magnitude x falls, say, between the boundaries 0 and 1, then it is not really the magnitude 1 but the zero point at which 1 is passed that constitutes the boundary, namely the terminal boundary. If we call the terminal boundary 0_2 and the initial boundary 0_1 , then 1 denotes that the distance between the two boundaries is posited as a magnitude determinate in full generality — that is, bounded — as a unit of distance. That x , in so far as it falls between 0 and 1, can with an unchanged initial boundary vary through countless terminal boundaries, or, as it is put, have infinitely many values, may then be expressed thus: $\frac{1}{\infty} = 0$, only by dividing the unit to infinity is

^{*)}“Everything that can be thought of as increased or diminished, and that can be resolved into parts of like kind, is called a magnitude (*quantum*) and is said to possess quantity (*quantitas*).” Ramus.

the real boundary reached; $\frac{1}{0} = \infty$, within every finite unit there falls an infinite number of boundaries; or $1 = \infty \cdot 0$, the finite unit arises through the boundary's being posited and annulled infinitely many times.

The determinate quantum presupposes a quantifying within qualitatively determining zero-boundaries. In virtue of the determining negations the finite magnitudes differ so far from one another that the difference can not only be stated but stated exactly: it is not enough that a is greater than b ; the question here is, how much greater? But since the difference of magnitude rests solely on fixed boundaries, and no finite boundary of magnitude stands so fast that it cannot — as ∞ reminds us — be annulled again, the finite magnitudes can be altered as the boundaries are altered: every magnitude can as magnitude be increased and diminished to infinity.

The dialectical unity of 0 and ∞ , which has its subsistence in the finite magnitude, $1 = \infty \cdot 0$, develops exactly through a constant alternation between the fixing of boundaries and the passing of boundaries; and so a decisive opposition forms between *fixed* and *fluid*, *being* and *becoming* boundaries. The immediate zero-boundary has no becoming; it comes to be in being posited. But since every boundary can be passed, and the alteration of boundary can continue so that the one boundary falls incessantly either inside or outside the other, the boundary can find itself in a becoming. The boundary of the boundary's becoming, the terminal boundary, may in turn be regarded partly as posited and partly as annulled: posited in so far as the magnitude is finite, annulled in so far as a finite magnitude may in its generality be as great and as small as one pleases. In positing a determinate terminal boundary, countless preceding terminal boundaries have already been passed, and so far annulled.

β) Transitions. In the boundary the bounded is reflected. To think the boundary without implying something bounded by it is as impossible as to grasp the meaning of a denial without thinking of what is thereby denied. The immediate zero-boundary is therefore posited straightway only on the presupposition that there is in intuition something given whose boundary it is. That such boundaries as points, lines and surfaces, although in and for themselves pure negations, are nevertheless *intuited* in positive form — the point as the line's smallest part, the line as the surface's edge, the surface as a side of the body — is a striking proof that the boundary goes together with the bounded. But since the boundary nevertheless distinguishes itself from the bounded as negation from the positive, and since this distinction is equally valid for thought and for intuition, there is contained in this the task of following what is to be bounded up to the boundary. How, then, does one get from boundary to boundary? By a motion! Here there is originally no

other way out, no other means. If one tries in thought to mark off on a line, say, two boundary points *A* and *B* at a certain distance, one must, if unconsciously, perform a motion, in that, to intuit the arbitrarily chosen distance, one involuntarily lets the eye run through the way from *A* to *B*. Let the eye fasten where it will, on its way it meets only boundary, for every point in the line is in and for itself a boundary point. As the eye moves along the line it looks as though it were merely following the moving boundary, the moving point; on this rests the mathematical explanation that the line arises by the motion of the point. Since this explanation rests essentially on an optical deception, it is no wonder that it turns out somewhat ambiguous. How can the pure point, the boundary, the abstractly negative, move? For that intuition, which in its immediacy presupposes the positive, can, once thought posits the negation, pass every boundary and go from boundary to boundary, is by no means yet sufficient to explain the mathematical miracle that the point, pure negation, should be able to move, to change place in space although it occupies no space, or as it were to stretch itself and form something so positive as a unit of length, a line. If the line could arise by a motion of the pure abstract zero point, then it would also have to be conceivable as composed of nothing but zero points, of $0 + 0 \dots + 0^*)$. To set boundary immediately beside boundary is impossible, for in abstracting from the bounded the two boundaries coincide; Herbart's "die starre Linie" is as much a logical as a mathematical monstrosity^{**)}; only in reflection of a positive can the negative come outside itself; only by means of something bounded can boundary follow upon boundary.

Between two successive boundaries, then, there is something bounded. If this bounded coincided immediately with the boundaries, so that there were nothing, nothing whatever, between, then the boundaries themselves would have to coincide and could not follow upon one another. If the bounded fell as an immediately existing thing between boundaries different from it, then there would be a finite distance, and hence countless boundaries between two immediately successive boundaries, which is a contradiction: all motion is grounded in the dialectical unity of the boundary and the bounded. By means of the boundary, motion is transition; the pure boundary is to that extent always a form of transition. It does indeed make an essential difference whether the transition is intuited in space or merely represented as an arithmetical transition from finite to 0, from finite to ∞ ; but this stands firm: without reflexes of space and time no motion,

^{*)}That this proposition by no means conflicts with the equation $\frac{a}{\infty} = 0$ taken as an expression of possibility becomes evident as soon as one recalls that the possibility depends precisely on 0 and ∞ being taken, not as empty negations, but as symbols of the greatest and smallest possible magnitude.

^{**)Cf. Den Herbartske Philosophie by P. S. V. Heegaard. Copenhagen 1861. pp. 21–45.}

without motion no quantifying, without quantifying no quantum, without quantum no mathematics. To exclude the category of motion from pure mathematics and still keep the categories "boundary," "transition," "approximation" is a contradiction. How the categories are chosen, stressed and applied in mathematics, so far as it is technically a question of the most suitable execution of exact operations, must of course be decided within mathematics itself; but the logical connection and dialectical transitions of the operational concepts are investigated in philosophy. If it then happens — for happen it may — that a specialist, a routineer whose mechanics of concepts has long since become one with his mechanics of operation, grows loud on such an occasion, and by a loud application of the Saracen dilemma expressly demands that philosophy give up dialectic and pay unconditional homage to the mechanics of concepts: such an occurrence cannot much interrupt the dialectical development, but rather hasten it.

If someone, in order to prove the proposition that "the transition from finite to finite coincides with the transition from finite to infinite," had appealed to an equation such as $\text{tg } 90^0 - \text{tg } (90^0 \pm \varepsilon) = 0$, where ε denotes a point of finitude, then a mathematician would certainly have had occasion here to point out the mistake, and to set the true equation against the false one: $\text{tg } 90^0 - \text{tg } (90^0 \pm \varepsilon) = \infty$. But when in a philosophical analysis (cf. *Phil. og Math.* pp. 23–24) there is no question at all of any special determination of the difference between $\text{tg } 90^0$ and $\text{tg } (90^0 \pm \varepsilon)$, but only of the well-known fact that $\text{tg } x$ passes from being finite to being infinite as x passes from a value less than 90^0 to 90^0 — then where does the mathematician (cf. *Begr. Nøiagt.* p. 47; "Afregning," revised, p. 7) mean to go with his $\text{tg } 90^0 - \text{tg } (90^0 \pm \varepsilon)$? If ε is of a finite value, however small, and hence a

point of finitude^{*)}, then $\text{tg } (90^0 \pm \varepsilon)$ is finite, and the transition from finite (90^0) through a point of finitude (ε) to finite ($90^0 \pm \varepsilon$) therefore coincides with a transition from infinite ($\text{tg } 90^0$) to finite ($\text{tg } (90^0 \pm \varepsilon)$). If ε is infinitely small, and hence a point

^{*)}One has, say, $\text{tg } 90^0 = \infty$, and one has $\text{tg } (90^0 \pm \varepsilon)$ finite; how does one now get from $\text{tg } 90^0$ to $\text{tg } (90^0 \pm \varepsilon)$? Plainly by ε , as part of the arc, being traversed continuously; and in the course of this motion it happens that the tangent passes from being finite to being infinite. How, then, is such a transition really to be thought? "It can," it is maintained, "happen only by a leap." What, then, is leapt over here? "A finite!" But then there is only a finite distance between finite and infinite. If nevertheless the transition from finite to infinite is to happen by a leap, then the transition from finite to finite must also, where a finite distance is concerned, happen by a leap; but if the transition through the finite can happen only by a leap, then all continuous transition, and with it every possible motion — indeed every possible transition — is denied. "An infinite!" But to arrive at the infinite by leaping over it is surely an intolerable and senseless contradiction. "Neither a finite nor an infinite!" But what sort of magnitude is it that is neither finite nor infinite? The greatest possible finite magnitude it cannot possibly be, for the greatest possible finite magnitude has precisely the property that by being *neither finite nor yet infinite* it is precisely at one and the same time *both* finite and infinite. "Nothing!" But what sort of leap is it, then, by which nothing is leapt over? (Cf. *Begr. Nøiagtighed* p. 45. The revised "Afregning" p. 7).

of infinity, then $\text{tg}(90^\circ \pm \varepsilon)$ is infinite, and the transition from finite through a point of infinity to finite (for $90^\circ \pm \varepsilon$ is surely of a finite value just as much as 90°) therefore coincides with the transition from infinite ($\text{tg } 90^\circ = \infty$) to infinite ($\text{tg}(90^\circ \pm \varepsilon) = \infty$). How great the difference between ∞ and ∞ may otherwise be does not concern the transition as transition^{*)}. The objection that the expression “coincide with” denotes merely “an accidental concurrence” is unwarranted; for changes of tangent do not merely occur *simultaneously* with, but are *necessary consequences* of, corresponding changes of angle: to confuse the consequence’s issuing from the presupposition with an accidental concurrence of something simultaneous betrays mere thoughtlessness.

^{*)}How infinitely dialectical the concept of magnitude is becomes evident when one asks, say: how great is the greatest possible finite magnitude? or: how small is the smallest possible finite magnitude? The demand expressed in these questions contains an essential contradiction. There is no finite magnitude so great that its boundary cannot be passed, so that a still greater is always possible; nor is there any finite magnitude so small that a smaller is not always possible. The greatest possible finite magnitude is, in other words, the infinitely great finite, and the smallest possible finite the infinitely small finite; here too it shows itself that the infinite lies, not in an impossibility, but in a possibility. How, then, can the contradiction be resolved? By seeing what is two-sided in the boundary of magnitude, that is, by taking the boundary partly as being and partly as becoming. The magnitude with a being boundary, the being magnitude, be it as great and as small as it will, is always finite; with a becoming boundary the magnitude, the becoming magnitude, moves either toward the greater, and then presents itself under the *infinite* becoming as the infinitely great, or toward the smaller, and is then grasped under the *infinite* becoming as the infinitely small: the infinitely great and the infinitely small are therefore magnitudes whose being boundary is resolved into becoming, as into unceasing growth and unceasing diminution. What is the notation for the greatest possible finite magnitude. $\lim x = \infty$! And for the smallest possible? $\lim x = 0$. The boundary that separates the finite from ∞ and from 0 is, so to speak, not a hair’s breadth wider than the one that separates the finite from the finite. “If the one point of finitude lies infinitely near the other, then the same may now be said of 0 and finite, and of ∞ and finite.” (Philos. og Math. p. 24). If one rejects dialectic and tries by the critical separation “either a finite or an infinite” to remove the contradiction, it only then becomes seriously insistent; and if one then tries by all manner of technical turns, leaps and artifices to veil the contradiction lurking everywhere, one may easily, by venturing now and then into the logical, get into several comical embarrassments.

If one holds fast to the dilemma “either ε is finite or it is 0” one-sidedly and critically, that is, undialectically — hence “ $\varepsilon = 0$ either finite or 0” — then one may as little say of 0 as of ∞ that it lies infinitely near the finite. If the mathematician will answer to this: “quite right! 0 does not lie infinitely near the finite either;” then one must ask again: what, then, lies between 0 and finite? The infinitely small the mathematician dare not adduce, for he has himself expressly said: “either ε is 0 or it is finite.” Nothing, then, lies between 0 and finite — and yet 0 is supposed not to lie infinitely near the finite! The case stands correspondingly with ∞ and finite. Undialectically one holds fast to the dilemma: either x is finite or it is infinite, *tertium non datur*; hence $\infty - x$ is either ∞ or finite or 0; and one may then as little say of ∞ as of 0 that it lies infinitely near the finite. But what, then, lies between finite and ∞ ? If the mathematician will answer “there lies ∞ !” he answers precisely the opposite of what he really meant. For he meant that, by the equation $\infty - a = \infty$, for $x = a$, there must absolutely be an infinite distance between the finite and ∞ — and behold, the supposed distance, the *what-lies-between*, turns into precisely that which was to be reached across the distance, namely the infinite itself. In other words: the outermost boundary of the finite cannot possibly be passed without the infinite being reached. What was to be kept infinitely far from the finite by “the infinite distance” thus comes infinitely near that finite after all. “Either finite or infinite!” But then nothing, nothing at all, lies between finite and infinite; and yet $\infty - a = \infty$, $\text{tg } 90^\circ - \text{tg}(90^\circ \pm \varepsilon) = \frac{1}{\cos 90^\circ} = \infty$ is supposed to be a proof, a fully valid mathematical proof, that between finite and infinite there is an infinite distance! If a mathematical critic should really win adherents by blustering with such logic, then on logic’s behalf one would indeed have to apply the old saying: *un sot trouve toujours un plus sot, qui l’admire*.

By $\frac{1}{a-x} = y$ the changes in y are point for point dependent on the changes in x ; and when y , as $x = a$, passes from being finite to being infinite, who will see in such a transition a merely external accidental concurrence with the value assigned to x ? If $\varepsilon = a - x$ is the smallest possible finite difference, then the turning point at which y passes from finite to infinite is also infinitely near; but from this it by no means follows that the difference between finite and ∞ can be parcelled out into either finite or infinite subtraction-differences. If the questions of continuity are to be decided by proofs from *immediate* subtraction-differences, the proofs will indeed turn out accordingly. The equation $\operatorname{tg} 90^\circ \div \operatorname{tg} (90^\circ \pm \varepsilon) = \frac{1}{\cos 90^\circ} = \infty$ holds also for $\varepsilon = 0$. By making the immediate subtraction-difference the measure of the continuous one arrives at the — in a logical sense — exceedingly beautiful result that, since $\operatorname{tg} 90^\circ - \operatorname{tg} (90^\circ \pm 0) = \frac{1}{\cos 90^\circ} = \infty$, there must necessarily be an infinite distance between $\operatorname{tg} 90^\circ$ and $\operatorname{tg} 90^\circ$, or, what here comes to the same, $\operatorname{tg} 90^\circ$ must be infinitely far from itself^{*)}. That ∞ , the in itself unbounded, can be a boundary rests essentially on 0, the determining negation, the boundary itself. That 0 lies infinitely near the finite means, in other words, that the boundary lies infinitely near the bounded; and what holds, with respect to the attainable boundary, of $a - x = 0$ in an equation such as $\frac{1}{a-x} = y$, holds also of $y = \infty$. The boundary ∞ is therefore, seen from this standpoint, as attainable as the boundary 0. From this it is in turn explained that, while no finite magnitude can denote any transition from + to –, ∞ can denote

^{*)}The evasion that ∞ is not a magnitude says nothing; if ∞ were not a magnitude, how could one have $\infty - \infty = n$, a determinate *difference of magnitude* between two *non-magnitudes*? What it means to say that ∞ is a magnitude, namely a becoming magnitude, is seen for instance from $\frac{1}{1-3} = 1+3+3^2+3^3 \dots + 3^n + \frac{3^{n+1}}{1-3} = \infty - \infty = -\frac{1}{2}$, for $n = \infty$. The evasion that $\operatorname{tg} 90^\circ - \operatorname{tg} (90^\circ \pm \varepsilon)$ is really a *function-difference* does not help either; for if the function-difference is to acquire significance as against the immediate subtraction-difference, ε must be transformed into an element, and thereby becoming, continuous change, transition — motion, that is — must be what decides.

One further remark for consideration. A proposition is set up, illuminated from several sides (cf. Phil. og Math. pp. 23–24, Begrebet Nøiagtighed pp. 45–47, “Afregn.” revised p. 7, Appendix p. 29), whose fundamental thought is this: in order to think the continuous transition from finite to ∞ it is necessary to look at the motion; if one abstracts from the motion and keeps to immediate subtraction-differences, the continuous transition becomes inconceivable. Against this a mathematician who is also an extraordinary logician comes forward with the announcement that the proposition set up is false, fundamentally false. How does the mathematical logician set about proving his claim? He demonstrates that the proposition’s mathematical error is “infinitely great,” in that by the equation

$$\operatorname{tg} 90^\circ - \operatorname{tg} (90^\circ \pm \varepsilon) = \frac{\sin (\mp \varepsilon)}{\cos 90^\circ \cos (90^\circ \pm \varepsilon)} = \frac{1}{\cos 90^\circ} = \infty$$

he expressly abstracts from the motion, and by abstracting from the motion proves clearly and incontestably what the assailed proposition precisely affirms: namely that the proof of a continuous transition must necessarily fail when one abstracts from the motion. The attempted refutation thereby turns into an express confirmation of what was to be refuted. So: one sets out to demonstrate a proposition’s “infinite mathematical error,” and behold, by a — of course, it goes without saying, insignificant — mathematico-logical blunder one comes precisely to prove the proposition’s mathematical validity: ordinary mathematics, extraordinary logic!

this transition just as well as 0, precisely because 0 can denote it. “Since a transition, as regards the opposition of positive and negative, acquires significance only through the positive’s passing over into the negative and conversely the negative into the positive — that is, through the positive and the negative both being in the transition and therefore belonging to the concept of the transition — such equations as $\sin 0 = 0$, $\cot. 0 = \infty$ ought strictly to be written $\sin (\pm 0) = \pm 0$, $\cot (\pm 0) = \pm \infty$, and so on. By this separation of + and – in 0 and ∞ there arises within these boundary notations themselves an inner difference which is dialectical in the highest degree. The notation ± 0 expresses, on closer inspection, that the one zero has divided itself into three different zeros: a +0, a –0, and the pure neutral 0, the sharp negation of + and $\div$. The same separation repeats itself with $\pm \infty$: here too there is a + ∞ , a – ∞ , and a neutral ∞ in which + and $\div$ have vanished. When, therefore, it is said that 0 and ∞ are notations of transition, this cannot hold of the pure 0 or the neutral ∞ , but only of ± 0 and $\pm \infty$ ^{*)}. The way in which the so-called indeterminate forms $\frac{0}{0}$, $0 \cdot \infty$, 0^0 , ∞^0 and 1^∞ are treated in mathematics shows plainly enough what forms of transition mean.” (Math. og Dialekt. p. 72).

y) *Articulations*. As forms of transition, then, 0 and ∞ denote the boundary’s dialectical unity with the bounded, and thereby motion: transition presupposes motion. In so far as the finite magnitudes are determined by given signs, they can, without any motion being noticed, be grasped by representation and held fast in their finitude — even to such a degree that it looks as though the infinite were only an empty abstraction, an external negation hovering above. But in so far as the finite magnitudes are determined by relations and grasped as concepts, the negation shows itself as dwelling within finitude, and the infinite is articulated. “So long as representation involuntarily clings to the sign 1, it looks as though the unit were an immediate something, an existing thing one could intuit and point straight at; this immediacy of representation disappears when one begins to grasp the unit as a relation: $\frac{2}{2} = \frac{3}{3} = \frac{4}{4}$.” (Mathem. og Dial. p. 79). What, then, is the unit? In and for itself every magnitude is the unit: $a = a$, $b = b$, $c = c$; as many different magnitudes, so many different units! But none of the different units is the unit itself; how, then, do we arrive at the unit that determines all units? By the equality expressed in equal magnitudes! From $\frac{a}{a} = 1$, $\frac{b}{b} = 1$, $\frac{c}{c} = 1$ it is expressly seen that the unit is precisely every magnitude’s equality with itself; and by specially bringing out the equality that holds for each magnitude, and positing this in turn as a magnitude, we have

^{*)}Only by regarding zero from 3 points of view — 1) as the starting point of the positive, 2) as the starting point of the negative, and 3) as the dividing point of unity of both — and letting these 3 determinations reflect themselves in one fundamental determination, does one think the concept of zero, the pure concept of boundary, as a concept of transition.

the general unit. As the determinate expression of equality, as the quotient of equal terms of a ratio, the unit is independent of the immediate difference of the terms; so long as each term merely continues to be equal to itself, the terms can be altered, can grow and diminish to infinity, without such alterations in the least altering the unit. If therefore one has, by means of the bounded ($a = a, b = b \dots$): $\frac{a}{a} + \frac{b}{b} + \frac{c}{c} + \dots = 1 + 1 + 1 + \dots$, then one has also, by means of the boundless ($\text{Lim. } \frac{x}{x} = \frac{\infty}{\infty}$): $\frac{\infty}{\infty} + \frac{\infty}{\infty} + \frac{\infty}{\infty} + \dots = 1 + 1 + 1 + \dots$ and, by means of the boundary ($\text{lim. } \frac{x}{x} = \frac{0}{0}$): $\frac{0}{0} + \frac{0}{0} + \frac{0}{0} + \dots = 1 + 1 + 1 + \dots$ Out of these streams of change the unit rises forth as the unchangeable, as quantifying's pure quantum, the immovable within motion.

On the unity of the general and the particular unit, the dialectical unity of the magnitude's concept and the magnitude's value, the function depends; every magnitude is a function of itself and the unit: $x = f(x) = 1.x$ (cf. Math. og Dial. pp. 6–7). From $\frac{f(x)}{x} = 1$, the expression for the magnitude's unity with itself in general, the *general unit* ($f(x) = x$), there follows $\frac{\int_{f(x)=0}^{f(x)=1.a} df(x)}{\int_{x=0}^{x=a} dx} = \frac{a}{a} = 1$, i.e. the magnitude's unity with itself as a particular magnitude, the *particular unit* ($f(a) = a$).

If the transition is presented, with $d.f(x) = dx$, by $f(x) = \int_{x=0}^{x=\infty} dx = \infty$, it must above all be held fast that there can here be no thought whatever of a direct summation, but only of becoming, coming-to-be, intellectual motion. Nothing has from the beginning done more to displace the decisive point of view and force back the natural development of the concept than the circumstance that the mechanics of operation has far too much treated the differential now as an immediate atom of magnitude, now as a mystical zero, instead of grasping it as a moment of transition, and has correspondingly regarded integration merely as a kind of summation (Math. og Dial. pp. 99–104) instead of determining it as a restitution of the finite through the infinite by means of the motion that holds for quantifying. How impossible it is to represent $\int dx = dx + dx \dots = \infty dx$ as a sum proper^{*)} is seen directly from the fact that dx , as a dialectical intermediate determination between 0 and x , cannot possibly be an addend. To be an addend, dx would have to have immediate finite existence — thus not only a but $\frac{a}{10000}$ can be an addend; but taken as existing, dx is a highly ambiguous magnitude, too great to be nothing, too small to be something. This ambiguity turns into a two-sidedness within the concept when dx is grasped as the determination of transition on which all motion essentially depends. As a determination of transition, dx cannot at any instant stand still; its being is becoming, it alters and becomes either 0 or x . If someone objects that one cannot get from dx to x by such a leap, but must first have $2dx, 3dx, \dots$ up to ndx ,

^{*)} x can indeed, seen under the aspect of infinite discreteness, be *thought as resolved* into $dx + dx + \dots$; but it is one thing to *think* infinite discreteness, another to *add* dx to dx — to infinity.

for $n = \infty$, it is not difficult to point out the conceptual confusion hidden beneath this.

For in order to articulate the motion it becomes necessary to distinguish between a single and a double becoming. In the single becoming, n in ndx is either finite or infinite. In so far as n is finite, ndx is an element, and as an element an ordinary dx ; if ndx is to become a finite magnitude, n must be ∞ . So long as the question here is still only of what is generally essential in the transition, we are looking throughout at the dialectical difference-unity of dx and x , and not at the difference between dx and ndx ; in the course of the motion every ndx is, as an element, on the point of becoming a finite, and as a finite of growing by an element. How great the element is cannot, in this comparison — the comparison being immediate — come into consideration, nor how small the finite magnitude is; the transition depends precisely on the greatest possible element and the smallest possible finite magnitude being reciprocal determinations. In the single becoming the motion goes from 0 to x not through myriads of dx but through dx . By representing the task contained in $\int dx = x$ as solved thus — $0 + dx + dx + \dots + dx = ndx = x$ — one entangles oneself in the fixed idea that the series must contain some particular dx whose place marked precisely the being boundary at which the sum of the elements first gave a finite magnitude. The difference between addition and motion is here striking. By adding dx to dx one never reaches x : $x - dx = x - 2dx = x - 3dx = x - ndx$, so long as n is finite, and under addition n is always finite; by motion, on the other hand, one passes immediately and instantaneously from dx to x , x being the smallest possible *finite* magnitude. The x into which motion transforms the dialectical dx is in turn dialectical, like motion itself; what is dialectical in x follows directly from the fact that every motion must begin before it can end. With the beginning of motion dx has immediately become an x , for motion in its immediacy is itself relation, reflection; but since the motion is still only at its beginning, x is the smallest possible. If the motion now stops just as what has begun was to be completed, then the finite falls back again into the element: the smallest possible x is then itself a dx . If on the other hand the beginning motion is completed, then the result is reached: the smallest possible x grows out beyond a smallest possible, takes root in the finite, and thereby becomes qualitatively different from its element. The x that arises from $\int dx = x$ therefore has, however small it may be (Math. og Dial. pp. 111–112), the completed motion for its presupposition; it is of a determinate finite value, can as a finite be divided into infinitely many infinitely small parts, and by this possibility produces the semblance of being a sum of countless dx .

The Eleatic objections to the possibility of motion^{*)} may in the present connection be traced to an undialectical identification of beginning and completion. For a motion to be completed it must of course have begun; if now the beginning motion is posited as identical with the completed one, then we have, as the cardinal point of the contradictions, this admittedly absurd proposition: "for a motion to be completed it must already beforehand have been completed." One demands, in other words, an absolute beginning; but to demand an absolute beginning is a self-contradiction, for no beginning is or can be absolute. In the course of motion every beginning is a continuation, and every continuation a new beginning. The dialectical unity of beginning and continuation is motion itself. Under the aspect of discreteness every motion therefore also shows itself as a sum of infinitely many motions. Seen from this standpoint there undeniably lie between x and dx infinitely many dx , and likewise between dx and 0 infinitely many $\text{Lim. } \frac{dx}{x}$; but to see how such discreteness is resolved into continuity we must consider particularly the doubled becoming, the becoming in which motion determines motion.

If we have $f(x) = y = ax$, and hence (Math. og Dial. pp. 84–85) $dy = adx$, then the relation between dy and dx points to a double becoming: a becoming of x by the increment $dx = dx$, and a becoming determined thereby, under which the y dependent on x grows by the increment $dy = adx$. Although dy is an element, and as an element an infinitely small value, vanishing in comparison with any finite magnitude whatever, just as dx is, it is nevertheless seen from this that the one element can be a times greater than the other; if for example $a = 2$, then $dy = 2dx$ and $dx = \frac{1}{2}dy$; if $a = 3$, then $dy = 3dx$ and $dx = \frac{1}{3}dy$, and so forth. That no immediate comparison can, however, be made between the elements themselves, and that nothing can here be formed out of such elements either by piecing-together or by parcelling-out, is easy to see. Instead of deriving $dy - dx = dx$ from $dy = 2dx$, one gives up subtraction as an unfruitful symbolism, applies division, and obtains $\frac{dy}{dx} = 2$, the differential coefficient, the factor by which dx must be multiplied in order to give dy (Math. og Dial. p. 81). What decisive significance it has for the articulation of the infinite by the finite that multiplication and division take the place of addition and subtraction can be shown without difficulty. The magnitude of a finite unit of time can be stated immediately, and so can the magnitude of the space traversed with a certain finite velocity in a finite time; and since the magnitudes are each immediately given, the comparison between them can also be made in so immediate a way that the terms themselves preserve a certain independence over against the relation in which they stand to one another. If the velocity on one track is 4 miles and on another

^{*)}Cf. Math. og Dialekt. p. 131 and elsewhere.

8 miles an hour, then the subtraction-difference thus given falls out as of itself. It is otherwise when elements of time and space take the place of the finite units of time and space. How long, after all, is an instant? That cannot be stated immediately, and just as little can it be stated immediately how long a stretch of road can be traversed at a given finite velocity in a single instant. Instead of remaining in an external relation, the elements must transform themselves into moments in a pervasive reflection; and for this multiplication and division, product and quotient, are needed. In product and quotient the terms have already given up their external existence, indifferent to one another, and transformed themselves into reflected moments of relation. So long as product and quotient are formed only from finite magnitudes, however, the corresponding reflection is still limited by the merely representational grasp^{*)}; but when the elements take the place of the finite magnitudes, the last remnant of representation's immediacy must vanish, and the concept — the pure concept made plain by the articulation — must hold. If we let $s = 4t$, $ds = 4dt$, $\frac{s}{t} = \frac{ds}{dt} = 4$ be different expressions for one and the same velocity, namely 4 miles an hour, then it is evident that, although the velocity is the same, the expression of velocity $\frac{ds}{dt}$ stands in quite another relation to thinking than the expression $\frac{s}{t}$. The quotient $\frac{s}{t}$ still keeps within the relation supported by representation, but in $\frac{ds}{dt}$ the whole insight rests on pervasive reflection; in the relation determined by them, ds and dt are nothing, nothing whatever in and for themselves, but reflexes of one another, and in this transparency determined by one another, so that they afford representation no finite foothold at all.

The becoming doubled within itself is articulated still more richly when the velocity answering to the motion changes at every instant. For a motion may, as regards velocity, be subject to infinitely many changes and yet, provided none of these changes occurs suddenly, show itself continuous. The continuity is conditioned by all the changes being subject to one and the same higher law and thus standing in a relation of mutual dependence.

“In so far as s is a space already traversed in the time t , one cannot yet know whether this space has been traversed with variable or with invariable velocity. To determine the velocity one must give the time an increment, that is to say: provide time for continued motion. If the increment to the time is called h , then, since $s = f(t)$, one has the series:

$$f(t + h) = s + \frac{ds}{dt} \frac{h}{1} + \frac{d^2s}{dt^2} \frac{h^2}{1.2} + \frac{d^3s}{dt^3} \frac{h^3}{1.2.3} + \dots$$

^{*)}To finite reflection, to merely representational thinking, it does indeed look as though the sum $a + b = (a + b)$ were just as reflected as the product $a.b = ab$, and the difference $a - b = c$ just as reflected as the quotient $\frac{a}{b} = c$.

If the increment h is a finite magnitude and the velocity constant, the whole series will reduce to $f(t + h) = s + \frac{ds}{dt} \frac{h}{1}$; if h is infinitely small, equal to dt , the same will hold even when the velocity is variable. When h is finite, on the other hand, the terms of the series can be increased to infinity. Seen from the standpoint of the concept, the series states a totality of objective determinations of reflection. The individual terms may, where s means the space, v the velocity and φ the force, be rendered by different corresponding expressions:

$$\begin{aligned} f(t + h) &= s + \frac{ds}{dt} \frac{h}{1} + \frac{d^2s}{dt^2} \frac{h^2}{1.2} + \frac{d^3s}{dt^3} \frac{h^3}{1.2.3} + \dots \\ &= .. + v \frac{h}{1} + \frac{dv}{dt} \frac{h^2}{1.2} + \frac{d^2v}{dt^2} \frac{h^3}{1.2.3} + \dots \\ &= .. + \dots + \varphi \frac{h^2}{1.2} + \frac{d.\varphi}{dt} \frac{h^3}{1.2.3} + \dots \end{aligned}$$

The categories *motion*, *velocity*, *force* and *change in force* are thus determined by objective reflection^{*)}. Immediately s denotes a length of space; but in the equation $s = f(t)$ this length of space is seen in the reflection of a length of time; it owes its existence to time, in so far as it is traversed in a given time. Immediately v denotes a velocity, but in the expression $\frac{ds}{dt}$ the concept of velocity is determined by the objective reflection of space and time in the given relation. Immediately force is denoted by φ , next by the reflection of velocity and time in the determinate relation $\frac{dv}{dt}$, and finally, as $v = \frac{ds}{dt}$ is resolved into the reflection

of space and time, by a doubled reflection in the relation $\frac{d^2s}{dt^2}$. In this way the changes of force can, with respect to the relation of space and time, be determined as reflections of reflections to infinity." (Phil. Propæd. pp. 165–167).

^{*)}To what degree an otherwise developed reflection may fail in developing an operational concept when the hints that exact analysis gives are entirely overlooked, Hegel has shown in his speculative interpretation of the motion of falling bodies (cf. Phil. Propæd. pp. 161–167). The well-known proposition that the spaces traversed with uniformly increasing velocity are as the squares of the times employed, Hegel interprets so that the quotient $\frac{s}{t^2}$ becomes a dialectical consequence of the abstractly *a priori* relation of the concepts of space and time: "Die der Einheit, als der Form der Zeit, entgegengesetzte Form des Außereinander des Raums, und zwar ohne daß irgend eine andere Bestimmtheit sich einmischt, ist das Quadrat: die Größe außer sich kommend in eine zweite Dimension sich setzend, sich somit vermehrend, aber nach keiner andern als ihrer eignen Bestimmtheit — diesem Erweitern sich selbst zur Grenze machend, und in ihrem Anderswerden so sich nur auf sich beziehend" [the form of the asunderness of space, opposed to unity as the form of time — and that without any other determinacy mixing itself in — is the square: magnitude coming outside itself, positing itself in a second dimension, thus increasing itself, but according to no determinacy other than its own — making this expansion its own limit, and in its becoming-other thus relating only to itself]. It escapes Hegel's attention in this case that, even if motion in full generality can be deduced from the relation between pure space and pure time — a deduction which must always be conditioned by supporting subsidiary representations — the square of the time cannot possibly arise without a varying velocity, and a varying velocity not possibly without a corresponding cause: here a constant force acting on matter moving in space according to the law of inertia.

The dialectical reciprocal determination of the concepts of continuity and discreteness now comes to light in the way the continuous motion is articulated, in that the elements themselves can not only be divided and divided again, but under given conditions even be resolved into elements of higher and ever higher order^{*)}. If the space s could not, say, be divided according to possibility into nothing but $ds = ds$, as the time t into nothing but $dt = dt$, then continuity itself, and consequently every continuous motion, would be inconceivable, and the relation between rest and motion, between faster and slower motion, entirely unintelligible.

“The difference between slow and fast motion must plainly rest on the first having more, the second less rest in it, and so on the first being nearer to rest than the second; a motion with an infinitely small velocity is precisely so infinitely near rest that it may be regarded as rest^{*)}. Nonetheless the opposition between rest and motion is qualitative: what rests does not move, what moves does not rest. If we now reject infinite magnitudes, the difference of velocity can be explained only by pure motion’s being interrupted at finite intervals by rest. That the body on A traverses only s in the unit of time t , while the body on B traverses $s_1 = 8s$, must then be explained by saying that the former body moves only for $\frac{1}{8}t$ and rests for $\frac{7}{8}t$. So long as the unit of time in which such an alternation takes place is an hour, the alternation of rest with motion is conspicuous; if the unit of time is 1 second, the alternation is less striking and looks like a jerky motion; in a unit of time of $\frac{1}{100}$ second the alternation will not be noticed at all, and the phenomenon will show itself as a smooth motion ... If the motion on A is to be continuous, time and space

^{*)}To reason thus — “the magnitude y may be thought of as divided into infinitely many dy , granted! but since dy thereby becomes itself the smallest possible part of a magnitude, dy cannot possibly be itself divisible, since by dividing dy one would get parts smaller than the smallest possible” — will not do; for $dy = a dx$, $dx = \frac{dy}{a}$, and, if we choose the function $y = f(x) = x^2$, $dy = 2x dx$, $dx = \frac{dy}{2x}$, shows plainly enough what it means to say that dy is divisible. But if an element of the first order is not absolutely indivisible, although as an element it is *infinitely small*, then it follows of itself that elements of the second order d^2y , the third order d^3y , the 4th order d^4y , the n^{th} order d^ny cannot on account of their *infinite smallness* be absolutely indivisible either. One may indeed in full generality set $\frac{d^2y}{dx} = 0$, $\frac{d^3y}{dx^2} = 0$, $\frac{d^4y}{dx^3} = 0$, $\frac{d^ny}{dx^{n-1}} = 0$, and, what is equivalent to this, $\frac{dx}{d^2y} = \infty$, $\frac{dx^2}{d^3y} = \infty$, $\frac{dx^3}{d^4y} = \infty$, $\frac{dx^{n-1}}{d^ny} = \infty$, denoting by 0 and ∞ that elements of different order are “*incomparable*” (cf. Math. og Dialekt. p. 96); but from this it of course by no means follows that $\frac{d^2y}{dx^2}$ should be smaller than $\frac{dy}{dx}$, or $\frac{d^ny}{dx^n}$ smaller than $\frac{d^{n-1}y}{dx^{n-1}}$; on the contrary! The relations may even entail $\frac{d^4y}{dx^4} > \frac{d^3y}{dx^3}$, $\frac{d^3y}{dx^3} > \frac{d^2y}{dx^2}$, $\frac{d^2y}{dx^2} > \frac{dy}{dx}$, $\frac{d^ny}{dx^n} > \frac{d^{n-1}y}{dx^{n-1}}$. The greater the quotient, into so many the more parts must the dividend be divided: the character of the dividends is determined by the measures chosen as divisors. Between the elements dy and d^2y , d^2y and d^3y and so on, in the quotients cited, there is here no immediately determined relation; the relation determined by $y = f(x)$ first arises through the measures dx^2 , dx^3 , dx^4 ... all being determined by $dx = dx$. Thus if $\frac{dy}{dx}$ is constant, say = a , then also $\frac{dy}{dx} = \frac{f dy}{f dx} = \frac{y}{x} = a$, and conversely. If $\frac{d^2y}{dx^2}$ is constant, say = $2a$, then also $\frac{f f dy}{f f dx^2} = \frac{f dy}{f x dx} = \frac{y}{\frac{1}{2}x^2} = 2a$, and conversely; if $\frac{d^3y}{dx^3}$ is constant, say = $6a$, then $\frac{f f f dy}{f f f dx^3} = \frac{f f d^2y}{f f x dx^2} = \frac{f dy}{f \frac{1}{6}x^3} = \frac{y}{\frac{1}{6}x^3} = 6a$, and conversely. Such relations may be continued to infinity.

^{*)}Cf. Math. og Dialekt. pp. 55–56.

must be divisible to infinity; on this point Aristotle saw rightly. But what Aristotle did not see and could not see, because he lacked the fulcrum that only the analysis of the infinite can give, is that by this infinite divisibility of time and space there are formed precisely infinitely small magnitudes differing from one another in such a way that they enter into determinate finite ratios with one another." (Math. og Dialekt. pp. 144–45). To the constant velocity $v = a = \frac{ds}{dt}$ there answers a single continuity — a continuity which, since it is marked precisely by the differential coefficient of the first order being constant, we shall here call *the continuity of the first order*.

If the single continuity can be conceived only through a corresponding single discreteness going to infinity, then the continuity doubled within itself cannot be conceived either without a corresponding doubling of discreteness. If the velocity changed at every instant gives us the series $v_1, v_2, \dots v_{n-1}, v_n$ in n successive instants, so that $v_p - v_{p-1} = dv$, whereby the increment of space ds in each dt is successively transformed into $ds_1, ds_2 \dots ds_n$, then here, since $v_n > v_{n-1}$ and $ds_n > ds_{n-1}$, there is in these two series, in so far as they are regarded *immediately*, no continuity to be discovered. Continuity, which demands that the same must repeat itself without interruption at every instant, shows itself only when we discover, in the velocity as in the space, the possibility of an element unchanging under the changes. Thus if it is given that the force φ is constant, then the change of velocity dv effected by it at each instant, $dt = dt$, is constant too; but the constant element of space answering to this is not ds ; it is, by $dv dt = d^2s$, precisely the element d^2s .) The continuity articulated by the functional law $s = at^2$, which presupposes that the differential coefficient of the second order is constant, we shall accordingly call a *continuity of the second order*.

If the doubled continuity demands a doubled discreteness, then in like manner the threefold continuity — that, namely, which answers to $s = at^3$ — demands a threefold discreteness, an element of space of the third order. For here it is not the force itself, φ , but the cause of the change of force, ψ , that is constant. The change of force, $d\varphi$, is then of course constant too. The constant $d\varphi$ now effects at each instant, dt , a constant

)By applying the summation symbolism one is then also led, through an immediate summation of d^2s , to the expression for s as a function of the time.

In the 1st dt the velocity is dv , and effects the change of space d^2s

in the 2nd dt $2dv$ $2d^2s$

in the 3rd dt $3dv$ $3d^2s$

in the n^{th} dt ndv nd^2s

After the lapse of the time t there has here been traversed, since $t = ndt$, $n = \frac{t}{dt}$, a space: $(1 + 2 + 3 \dots + n)d^2s = \frac{1}{2}n(n+1)d^2s = \frac{t}{dt} \left(1 + \frac{t}{dt}\right) \frac{d^2s}{2}$, which, since $\frac{t}{dt}$ is infinitely small in comparison with $\left(\frac{t}{dt}\right)^2$, is equal to $\frac{t^2}{2} \frac{d^2s}{dt^2} = \frac{gt^2}{2}$, the usual expression for the law of falling bodies.

change in the increment of velocity, which in turn effects a constant change, $d^3s = d^3s$, in the increment of the increment of space; d^3s is therefore, in space, the element of discreteness answering to this law of continuity. It is the third differential coefficient, $\frac{d^3s}{dt^3}$, that is here constant. *The continuity is of the third order.*

From the articulations thus exhibited it is seen that continuity is by no means interrupted by more and ever more changes being drawn under the law of continuity. If continuity were ever to burst under the weight of the multitude of changes, then an insurmountable barrier would at last have to show itself to the rising order of the elements, without this proceeding from any constant differential coefficient; but since this is a contradiction, the order of the elements being able to rise and rise to infinity, the continuity of the n^{th} order answering to a constant n^{th} differential coefficient, $\frac{d^n s}{dt^n} = C$, must hold for $n = \infty$ as well^{*)}.

If we gather the results here won into a single survey, the following main points are especially to be brought out: 1) the circumstance that between dx and x there may lie countless dx , and, in so far as x is a dependent variable, between 0 and dx countless d^2x , between 0 and d^2x countless d^3x and so on, does indeed indicate that between the corresponding and simultaneously vanishing increments of the dependent and the independent variable there are varying ratios, from which differentials of higher order necessarily follow; but by no means that 0, dx and x (the smallest finite) are, as far as the concept of transition in its generality is concerned, separated by countless intermediate terms. When dx is to represent the smallest possible magnitude in an immediacy answering to independence, it cannot possibly become smaller without becoming 0, nor greater without becoming x . In its *immediacy* the element is an outermost boundary of discreteness, an atom of magnitude, and as such indivisible; reflected in its relations to other elements it may on the contrary show itself infinitely divisible. Motion depends on the dialectic of the element indivisible-divisible in itself. 2) The beginning of motion takes place immediately neither with 0, nor with x by itself, nor with dx , but with a negative-concrete unity of all three moments. "In the absolute Now nothing is and nothing happens. In the relative Now the infinite is posited as the beginning of the finite: the beginning of the finite therefore coincides with, is in the mathematical instant simultaneous with, the beginning of the infinite. In the empirical instant the finite is posited as completed; but in and with this completion of the finite an infinity is itself completed: the completion

^{*)}If s is a fractional, irrational or transcendent function of t , say $s = \cos t$, then there is no constant differential coefficient here; but this by no means means that continuity is broken — on the contrary, that it is so infinitely articulated that one cannot exhibit in isolation the atom of discreteness answering to such a continuity.

of the finite therefore coincides with, is in every actual instant simultaneous with, the completion of an infinity." (Math. og Dialekt. pp. 134–35). 3) Although motions of every kind must always presuppose at least subsidiary representations, reflexes, of time and space, motion is nevertheless in the strict mathematical sense to be confined to pure quantifying. And since a quantifying as necessarily presupposes a quantifying quantum as motion presupposes a something that can move, while yet the abstract quantum which furnishes quantifying's immediate presupposition owes its existence precisely to the quantifying, and thereby comes to presuppose that of which it was itself to be the presupposition — it is seen from this that quantity as an abstract concept can first acquire its true significance in and through an existing thing whose quantity it is.

Of pure mathematics it is said that it finds "application" to actual objects (in mechanics, astronomy, optics and the like) Essentially understood, such an "application" presupposes that the mathematical is originally realized in the objects to which mathematics is applied. An application of mathematics to actual objects would be, if not a sheer impossibility, at any rate a subjective arbitrariness, if the mathematical did not itself belong with inner objective necessity to the essence of actuality; but as a moment in the essence of actuality the mathematical belongs also to philosophy.

B.

Philosophy's Relation to Chemistry:

The Problem of Change.

In order to present philosophy's relation to chemistry, it will in the present connection be necessary: a) to characterize chemistry as a distinct science; b) to show how the problem of change arises in the course of chemical analysis; and finally c) how philosophy appropriates the problem developed within chemistry.

That philosophy, however deeply it immerses itself in an other, nevertheless relates itself through this other to itself, and so far moves within its own domain, will in the course of the exposition be recognizable in this: that nothing of the nature of material is taken up immediately from outside, but all positive constituents are so appropriated that they yield, each in its own way, ideal contributions to the knowledge of essence and serve the dialectic of the development of concepts.

a) Chemistry as a Distinct Science.

“Since science has its root in human life, where the ideals of imagination go before those of thought, it is natural that the various sciences should from the first come into the world in the swaddling-bands of fancy and be encumbered with unscientific admixtures. Thus astronomy was long grown together with astrology, chemistry with alchemy, philosophy with mythology. It does indeed lie in science’s own drive to throw off the chrysalis, to break its husk and purge away the foreign constituents. But cultivation costs time and purging costs effort; the hindrances the human spirit has here to overcome are of various kinds.” (Phil. Propæd. p. 70).

It is science’s task to penetrate into the interior of things and through the phenomenon to discover the essence; but recognized as this demand is in abstract generality, its fulfilment is correspondingly difficult when attention turns to the particular, the concrete, the actual. What phenomenon is better known or more conspicuous than the phenomenon of fire — and yet, what is the inner in this outer, what is the essence of flame, what is fire? Most people will at least have heard in passing that in recent times a number of metallic substances formerly unknown have been isolated by art; among these substances is potassium, a tin-white, brightly lustrous, very soft and pliable metal. If a piece of this metal is thrown into water it does not sink to the bottom but moves about on the surface, takes fire and burns with a violet flame; after which it seems to disappear entirely. What has happened here? Is it the potassium that has burnt the water, or the water that has burnt the potassium? *What, in the phenomenon of combustion, is that which burns, and what is that which is combustible?* To answer such a question science has had to prepare itself for some 3000 years — if, to begin at the beginning, we look for our first chemists among the priests of Egypt.

α) The Origin of Chemistry^{*)}.

Among the Egyptians the cultivation of chemistry seems to have formed part of the mysteries hidden in the temples; among the Greeks the elements of chemical knowledge passed, like other empirical knowledge, into philosophy; in the Middle Ages chemistry as alchemy was a direction for fathoming the riddle of nature and finding the philosophers’ stone (*lapis philosophorum*).

Since insight into the connection of natural forces is of no less practical than theoretical interest, and depends as much on rich experience as on sustained reflection, it is

^{*)}*Leçons sur la philosophie chimique par Dumas, recueillies par Bineau. Bruxelles 1839. Geschichte der Chemie von Dr. H. Kopp. 4 Bd. Braunschweig 1843–47.*

intelligible that the study of chemistry should long have been grown together with various alien endeavours, and that this branch of science should long have served as a means before it could be set free to be its own end and be developed in a purely scientific spirit. *Mystical speculation, groping after method, and the discovery of method* are then the stages that here mark the path of the development. *αα) Mystical Speculation: Antiquity and the Middle Ages.* Without experience, no thinking; but when visible things are only fleetingly surveyed, because the art of observation is not yet developed, abstracting thought unites with imagination and forms out of a few accidental and unreliable experiences a system of concepts whose content, so far as existing things are concerned, belongs more to fancy than to actuality. Conceptual forms can indeed, as Greek philosophy sufficiently shows from so many sides, be developed in abstract directions with clarity and sharp consistency although the content of experience is lacking; but if concepts of existence, the real concepts proper, are in question, it is self-evident that the development of form must everywhere be conditioned by a corresponding development of content.

The way in which Aristotle grounds the existence of four elements shows precisely how impossible it is to determine a thing's essence without knowing the thing. The elementary qualities — hot, cold, dry and moist — are brought out in such a way that each of them is posited as an essentiality in its own right, a substantial form determining matter; thus a simultaneous union of dryness and heat gives *the element of fire*, of heat and moisture *the element of air*, of moisture and cold *the element of water*, of cold and dryness *the element of earth*. It is a remarkable confusion of effect with cause, of the derived with the original, that we meet here. The thing itself, the existent, does not after all arise through the union of qualities, and so not through the union of fundamental qualities either; on the contrary, the union of qualities — and hence of fundamental qualities too — is posited in and with and by means of the thing. What we have here is not a system of general qualities subsisting on their own and acting upon things, but a system of things acting reciprocally upon one another through their qualities; it is not, for example, the quality dryness that dries the thing, any more than it is the quality heat that warms it; the cause of heat and of dryness is to be sought, not in a general essentiality, but in a conditioned action. What influence such a misdirection, inherited from philosophy and especially from Aristotle, must have exercised on the Middle Ages — fermenting as they were with deep intimations and obscure but bold endeavours — is not, so far as chemistry in particular is concerned, difficult to imagine.

Material substances such as quicksilver and sulphur must, in so far as they belong among externally actual things, show themselves as objects of experience calling for

corresponding observation; but since speculation prefers to see in such substances only a sensuous embodiment of certain non-sensuous essentialities stamped upon matter, it came about by degrees that the alchemists' *Mercurius* and *Sulphur* denoted, not primarily the quicksilver and sulphur actually existing, but above all these substances' metaphysical characters. Once the thing, which by its nature ought to be an object of observation, has been transformed into a speculative essentiality, it lies near to regard the sensuous as a fleeting image of the supersensuous, the natural as a symbol of the spiritual. The black powder that arises when sulphur and quicksilver are ground together passes on ignition into a red mass, namely cinnabar; in one and the same substance red and black, the symbols of light and darkness, of the good and the evil principle, are therefore visible. On the presupposition that in all metals there was to be found *Mercurius*, the character of the permanent and the indissoluble, shining in the metallic lustre, and *Sulphur*, the combustible, volatile and changeable, and that the mutual difference of the metals depended chiefly on a peculiar mixture and "fixation" of these natures, the alchemists were carried ever further in their endeavour to transform, transmute and ennoble the metals.

"What was then regarded as the magicians' highest scientific striving aimed at purifying the divinely-homogeneous which was found in sinful nature, bound up with resisting elements, so that what lay hidden as the divine and sustaining might work freely in every inward form. This is alchemy — no accidental, arbitrarily contrived thing, but a wholly essential constituent of the prevailing physics. All physicists sought the philosophers' stone, for there was then no other physics, and no other could arise. The bringing forth of this noblest kernel of all existence was as much of a religious kind as it was a physical experiment, and this universally prevailing endeavour gives the most striking proof of the spirit's bondage to the earthly. What was thus purified, in which the original creative force concentrated itself, must, when applied to the macrocosm, bring forth the noblest materials, precious stones and above all gold; but applied to the microcosm it must on the same ground — the original sustaining principle in both being the same — promote health. and prolong life." (H. Steffens)*).

Mystical, marvellous and enigmatic that substance must indeed have been which by mere contact was to subdue *Sulphur*, enchant *Mercurius*, *Metallorum Draco*, and transmute every metal, even the basest, into pure gold. The philosophers' stone, the great elixir, *Magisterium magnum*, the red tincture, which has the power to ennoble the metals on so prodigious a scale — *mare tingerem, si mercurius esset*, says Raimundus

*Cf. "Naturvidenskabens Forhold til Tidsaldere og deres Philosophie", a critical review by H. C. Ørsted. Aanden i Naturen. 2. pp. 161–62.

Lullus — and to heal all diseases, is accordingly not formed as an aggregate but grows from within, organically, like a living being. No wonder that the discovery of such a secret came at last (Basilius Valentinus) to appear as much a religious as a scientific task. The stone, which shone with a light of grace as well as of nature, spread a magical gleam over the whole of existence; earthly life with its sorrows and sufferings was then, seen in this light, a "digestion," a purification by "fermentation," the grave the place where a "putrefaction" should work which destroyed only the base constituents of the human body, while what is immortal in the soul emerged as by a "sublimation" of the nobler essence, a *Quinta Essentia*.

ββ) Groping after Method: Qualitative Analysis. The alchemists' reveries pointed in two directions: the *metallurgical* and the *medicinal*. That the philosophers' stone, once discovered, would heal all diseases was of course a fancy that could not possibly yield a scientific task; yet beneath this fancy there smouldered a spark of something rational, a thought which, though in fantastic form, broke through in so-called iatrochemistry, medical chemistry: the thought that the organic functions depend on chemical processes. The chemists of the sixteenth and seventeenth centuries — Paracelsus, Van Helmont, Sylvius — were of course by no means yet able to form any clear concept either of the chemical processes or of the organic functions, and so not able either to devise any definite, reliable or consistent procedure. What a Sylvius explains about the abnormal predominance of acids and alkalis in the body's fluids is at bottom as untrustworthy as what a Paracelsus reveals to us in the proclamation that every part of the human body has its own peculiar sulphur, its own mercury and its own salt; that a preponderance of sulphur gives fever and plague, a preponderance of quicksilver melancholy and paralysis, an excess of salt dropsy, and so forth. Yet the new direction nonetheless gave the investigations a new impetus. To draw the "quintessence" out of medicinal plants, to prepare the many kinds of tinctures, essences, extracts, decoctions and juices — these were endeavours well fitted to sharpen the spirit of observation, to multiply experiments, and to teach the individual investigator to exercise his eye and his judgment. It lies in the nature of science that it can become conscious of its end only in the same degree as it can become conscious of the procedure answering to that end, the method, and conversely. The end of medical chemistry, though in and for itself rational, must always show itself from a more or less fantastic side so long as research is unable to provide the means, so long as the road of investigation that leads to the goal has not yet been found; but the dialectic of progress is that on the path of error too there are truths to be found — among others the truth that here is an error. By the time iatrochemistry had

been carried far enough for its hollowness and charlatanry to show themselves properly, chemistry itself, following the other track, the ancient metallurgical one, had come to the turning point at which it could at last stand forth independently and begin to operate on its own account.

Chemistry's emancipation is recognizable in the altered task; for the task is now in the main neither to transmute the metals into gold nor to find the arcanum of medicine, a universal remedy against all diseases, but to investigate the conditions and laws governing the separation and combination of substances, and above all to answer the question: on what does combustion depend? But scientific, and distinctively chemical, as this task presents itself, it was impossible that the first attempts at its solution should turn out other than somewhat doubtful. The presupposition inherited from the alchemists, that in all metals there was a hypothetical elementary substance conditioning combustion, still continued to hold; nor could the prejudice easily be shaken off that combustion must absolutely be a destruction, a dissolution, a separation, so that something departed from the burning body, namely what went up in the flame. In this, however, the problem of combustion differs essentially from the iatrochemical fancies: that however much of the fantastic mingles with it at the outset, it lies so near to research that by its own rational character it can gradually point the way and lead into the method.

The changes that all combustible bodies, organic as well as inorganic, undergo in fire, Stahl^{*)} brought together under a single generally valid point of view, deriving combustion from a substance dwelling in all bodies, a substance of combustibility, namely phlogiston. Thus was formed the doctrine of phlogiston characteristic of the period from the latter half of the seventeenth to near the close of the eighteenth century — Stahl, Priestley — the phlogistic theory. By testing the analogies, bringing out the general, drawing theoretical inferences from the observation of phenomena, and applying more comprehensive rational points of view, science gathered its forces and so rose step by step into its own distinctive shape.

But despite the long series of experiments that the question of combustion alone must call forth, despite the multitude of discussions that the hypothetical substance phlogiston must occasion — whether it was sulphur, which Stahl denied; whether it was not rather a matter of light; or whether, as others found plausible, it might perhaps be hydrogen[†] — discussions which for all their hovering were always distinguished

^{*)}Georg Ernst Stahl, born at Anspach in 1660, was in 1687 appointed physician in ordinary to the Duke of Weimar, and, at Friedr. Hofmann's instance, in 1694 professor of medicine at the University of Halle.

[†]*Translator's note.* Danish *Brint* for hydrogen and *Itt* for oxygen (below, p. 91, with *Metaliliter* for metallic oxides) are native coinages of H. C. Ørsted, who appears in the footnote on p. 86; they replaced the international *Hydrogen* and *Oxygen* in Danish scientific prose and are still the ordinary Danish words.

from empty speculation by the fact that they all turned on actual matters of fact and corresponding presuppositions: the investigations were nevertheless still one-sided and imperfect so long as emphasis was laid merely on the quality of the given constituents, without the quantitative in the combinations coming particularly into consideration. So, despite the many and even connected special investigations, things remained on the whole with the old abstraction: that since like effects must have like causes, there must underlie the prominent properties of the various bodies, the ruling qualities, a corresponding common element of property or quality – in all acids a quality of acidity, an original acid (*acidum primitivum, primordiale, catholicum*); in all earths an earthy quality, an original earth; in all caustic alkalis a caustic quality, a caustic principle; in all combustible bodies a principle of combustibility, a phlogiston.

γγ) *The Fixing of Method: Quantitative Analysis.* Already in antiquity attention was drawn to the difference the metals display when tested in fire, some – such as gold – keeping their properties unchanged, while others, such as copper and iron, are transformed into earthy bodies. What remains when the metal has been burnt, the Greeks and Romans denoted by the same name as holds for burnt wood, namely ash (σποδός, *cinis*); but when the Arabs later brought out the resemblance between metallic ash and burnt lime, there arose in the West such expressions as *Calces*, metallic calxes (answering to what are now called metallic oxides), and *Calcinatio*, calcination, the condition into which the metal comes by combustion.

How closely the phlogistic theory attaches to the view that the metals are composite substances, and how the metallic calx accordingly stands to phlogiston and to the pure metal, may be shown briefly by an example. The more phlogiston a body contains, the more combustible it is, and the less ash remains after combustion; carbon, which leaves very little ash, is therefore particularly rich in phlogiston. If now rusted – that is, burnt – iron, the calx of iron, is heated with charcoal, the carbon will give up its phlogiston to the calx; there will thus arise calx of iron with phlogiston, and dephlogisticated charcoal; and the calx of iron in its union with phlogiston will constitute the reduced metal, pure iron, a composite substance.

What is distinctive in this emerges still more clearly when the antiphlogistic theory is set sharply against the phlogistic. The metals – iron, for example – are, according to the antiphlogistic theory, not composite but simple substances; combustion takes place not by something departing but on the contrary by something being added; it consists, in other words, not in a separating but in a combining. Pure iron burns in uniting with

Nothing in the argument turns on them, but the reader should know that Nielsen is using a deliberately Danish chemical vocabulary here.

oxygen; oxidized iron is reduced, not by its taking up phlogiston — for phlogiston exists only in fancy — but by its losing again the oxygen taken up, and being presented in its simplicity as an element.

Now that the two theories, each according to its fundamental thought, have been made plain by their mutual opposition, the question presses itself upon us of how we are to convince ourselves of the truth of the one and the untruth of the other — a question on which so much the greater weight must be laid, since an insight into what really determines chemistry's method depends on its being answered. If one weighs the iron before and after its combustion, one easily convinces oneself that the burnt mass weighs more than the unburnt: with the recognition of the significance lying in such a change of weight, quantitative analysis begins; and once quantitative analysis is combined with qualitative, it is all over with the phlogistians, for now, but only now, can an all-round examination of the phenomena begin.

What, after all, is phlogiston? If phlogiston is the combustible, what then is that which feeds the fire? When a body increases in weight by combustion, how is it thinkable that combustion can depend on a secretion of phlogiston? How can a body become heavier by losing something of its mass? How does it come about that, when a body burns in atmospheric air, not only is a constituent of this air absorbed, but the burnt body has gained precisely an increase in weight answering to the weight of the absorbed air? When the adherents of phlogiston came to answer these and similar questions exactly, punctually and clearly, they entangled themselves in so many contradictions that Lavoisier, who wrote on his balance *in hoc signo vinces*, won the completest victory.

In the particular the general reflects itself: the grounding of a true theory of combustion coincides on the whole with a grounding of the chemical method. That matter is constantly maintained through all operations so that no creation and no annihilation takes place; that the weight of every chemical product is always equal to the weight of its constituents; that every increase of weight accordingly denotes a combination and every decrease a separation; and that it thus becomes possible to sense the otherwise non-sensuous and to follow matter's exchanges along otherwise invisible paths — these are now the fundamental principles on which the method thus won will henceforth know how to build. "For centuries," says Liebig, "the physicists measured; it is only fifty years since they began to weigh. Lavoisier has the balance to thank for all his great discoveries — the balance, that incomparable instrument which holds fast all observations and discoveries, which conquers doubt and draws truth forth into the light ... With the balance Aristotle's realm came to an end; his method, of making the explanation of a

natural phenomenon a game for the mind, gave place to genuine investigation of nature."

β) Chemical Forces.

γ. Chemical Combinations.

b) The Problem of Change in Chemistry.

α) Atom and Equivalent.

β) Atomic Constitution and Crystal Form.

γ) Atom and Substance.

c) The Problem of Change in Philosophy.

α) The Atom as Point of Discretion.

β) The Atom as Point of Function.

γ) The Atom as the Substantial Moment.

Result.

C.

Philosophy's Relation to Physiology:

The Problem of Life.

a) The Essence of Life.

α) The Organism.

β. The Organic Type.

γ) The Organic Purpose.

b) The Stages of Life.

α) Plant Life.

β) Animal Life.

γ. Human Life.

c) Soul and Body.

α) Metaphysical Theories.

III.

Philosophy and Academic Study.

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